DESIGN, SIMULATION, AND IMPLEMENTATION OF A RECONFIGURABLE INTELLIGENT SURFACE (RIS) PROTOTYPE FOR WIRELESS SIGNAL ENHANCEMENT

A PROJECT REPORT

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by

NAMAN KAMBOJ	**23BEC1425**
SURIYA DHARSAUN	**23BEC1389**
ABHIGYAN SINGH NEGI	**23BLC1143**

Under the guidance of

Dr. Vigneswaran T
School of Electronics Engineering (SENSE)



SCHOOL OF ELECTRONICS ENGINEERING
VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Vandalur–Kelambakkam Road, Chennai – 600 127

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DECLARATION

We hereby declare that the report entitled *"Design, Simulation, and Implementation of a Reconfigurable Intelligent Surface (RIS) Prototype for Wireless Signal Enhancement"*, submitted by us to the School of Electronics Engineering, Vellore Institute of Technology, Chennai, in partial fulfilment of the requirements of the course BECE497J Project-I of the Bachelor of Technology programme, is a record of bonafide work carried out by us under the supervision of Dr. Vigneswaran T, School of Electronics Engineering.

We further declare that the work reported in this report has not been submitted, and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or in any other institute or university.

Place: Chennai
Date: _____

Naman Kamboj	Suriya Dharsaun	Abhigyan Singh Negi
23BEC1425	23BEC1389	23BLC1143

CERTIFICATE

This is to certify that the report entitled *"Design, Simulation, and Implementation of a Reconfigurable Intelligent Surface (RIS) Prototype for Wireless Signal Enhancement"* is prepared and submitted by Naman Kamboj (23BEC1425), Suriya Dharsaun (23BEC1389) and Abhigyan Singh Negi (23BLC1143) to Vellore Institute of Technology, Chennai, in partial fulfilment of the requirements of the course Project-I of the Bachelor of Technology programme, and is a bonafide record of work carried out under my guidance. The project fulfils the requirements as per the regulations of this Institute and, in my opinion, meets the necessary standards for submission. The contents of this report have not been submitted, and will not be submitted, either in part or in full, for the award of any other degree or diploma.

Signature of the Guide	Signature of the Head of Department
Dr. Vigneswaran T	Programme Chair
School of Electronics Engineering	School of Electronics Engineering
Date:	Date:

Internal Examiner	Internal Examiner
Date:	Date:

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Naman Kamboj
Suriya Dharsaun
Abhigyan Singh Negi

ABSTRACT

Indoor Wi-Fi communication is affected by signal attenuation, multipath fading, and coverage gaps. This project studies the design, simulation and experimental check of a reconfigurable intelligent surface (RIS) operating at 2.45 GHz, in real-world scenarios. The unit cell design simulation resulted in a reflection phase difference of 140.8° between the two diode states, whereas addition of 22 nH RF chokes into the bias network for practical design simulations resulted in a phase difference of 139.7°. A PIN-diode-switched unit cell was designed with FR4 substrate as base and was validated using Ansys HFSS. After the validation the design was scaled to a 4×4 RIS array steered which was tested and resulted in a beam steering of 10.0° and 18.1° for target angles of 10° and 20°, respectively. A simulated analysis was performed to evaluate the effects of array size, steering accuracy, and propagation distance. After the validation of the design, the unit-cell PCB was fabricated and assembled for testing using two ESP32 boards and an Arduino controller. The hardware testing measurements showed that introducing the RIS into the obstructed propagation path changed the mean received signal strength indicator (RSSI) across the tested distances. The measured RSSI increased by 1.25 dB at 60 cm, 5.27 dB at 1 m, and 3.15 dB at 3 m with the RIS in the OFF state, relative to the obstructed condition without the panel. With the RIS in the ON state, an improvement of 3.64 dB was observed at 1 m, while lower RSSI values were recorded at the other two distances. The project establishes a practical foundation for RIS-based wireless signal enhancement and identifies array size ,reflection losses, panel placement, and measurement repeatability as key areas for further investigation

Keywords: Reconfigurable Intelligent Surface, PIN diode, 1-bit coding, Floquet analysis, RF choke, Wi-Fi RSSI.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

Term	Meaning
AF	Array factor
BOM	Bill of materials
DC	Direct current
DCR	DC resistance of an inductor
EIRP	Effective isotropic radiated power
ESD	Electrostatic discharge
ETSI	European Telecommunications Standards Institute
FR4	Glass-epoxy laminate grade FR-4
GPIO	General-purpose input/output
HFSS	Ansys High Frequency Structure Simulator
HPBW	Half-power beamwidth
ISM	Industrial, scientific and medical (radio band)
LoS/NLoS	Line of sight / non-line of sight
MOSFET	Metal–oxide–semiconductor field-effect transistor
PCB	Printed circuit board
PIN	p-type/intrinsic/n-type (diode)
PRR	Packet reception rate
RCS	Radar cross section
RF	Radio frequency
RIS	Reconfigurable intelligent surface
RSSI	Received signal strength indicator
RX/TX	Receiver/transmitter
SDG	Sustainable Development Goal
SMD	Surface-mount device
SOD-323, SOT-23	Standard two-lead diode and three-lead transistor packages
SRF	Self-resonant frequency
STAR-RIS	Simultaneously transmitting and reflecting RIS
Wi-Fi	Wireless local-area networking based on IEEE 802.11

Symbols

Symbol	Meaning (unit)
c	Speed of light in vacuum, $299\,792\,458\,\mathrm{m\,s^{-1}}$
f_0	Design frequency, 2.45 GHz
λ_0	Free-space wavelength at f_0, 122.36 mm
k_0	Free-space wavenumber, $2\pi/\lambda_0$ ($\mathrm{rad\,m^{-1}}$)
Γ_{ON}, Γ_{OFF}	Complex reflection coefficient of the unit cell in each diode state
S_{11}, S_{21}	Reflection and transmission scattering parameters
$\Delta\phi$	Effective (wrapped) phase separation between the two states (°)
a, b	Common and switchable parts of the one-bit reflection coefficient
d	Centre-to-centre element pitch (mm)
N	Number of elements along one side of an array
θ	Angle measured from the surface normal (°)
ψ	Inter-element phase, $k_0 d \sin\theta$ (rad)
X_L, X_C	Inductive and capacitive reactance (Ω)
C_T, r_f, L_S	PIN-diode capacitance, forward resistance and package inductance
I_F, V_F	Diode forward current (mA) and forward voltage (V)
d_0, d_1, d_2	TX–RX, TX–RIS and RIS–RX distances (m)
r_i, $\bar{r}$, s_r	RSSI sample, sample mean and sample standard deviation (dBm, dB)
N_{tx}, N_{rx}	Packets transmitted and received

CODES AND STANDARDS

Table 1 lists the standards, regulations and technical reports that govern or inform the design, the planned experiment and the hardware handling in this project. None of the listed documents has been formally tested for compliance; they are used as design references.

Table 1. Codes, standards and regulations relevant to this project.

Document	**Subject**	**Use in this project**
IEEE Std 802.11-2024 [1]	Wireless LAN MAC and PHY	Channel plan and signal format of the ESP32 Wi-Fi link used for the RSSI experiment (channel n centred at $2407 + 5n$ MHz).
G.S.R. 45(E), 2005, Government of India [2]	Licence-exempt use of 2.4 GHz to 2.4835 GHz	Permits low-power operation in the band; the ESP32 transmit power (19.5 dBm typical maximum [3]) is far below the 1 W transmitter and 4 W radiated-power limits.
ETSI GR RIS 001 [4]	RIS use cases, deployment scenarios, requirements	Terminology; indoor coverage enhancement as a recognised RIS use case.
ETSI GR RIS 003 [5]	RIS communication and channel models	Framework for the system-level link model in Chapter 5.
ETSI GR RIS 004 [6]	RIS implementation and practical considerations	Hardware constraints: bias networks, element losses, control.
IEEE Std 149-2021 [7]	Antenna measurement practice	Guidance for future pattern and gain measurements of the fabricated surface.
IPC-7351B [8]	SMD land-pattern design	Reference for checking the choke and diode land patterns against the purchased parts (Section 6.2.1).
ANSI/ESD S20.20-2021 [9]	ESD control	Handling of the PIN diodes and MOSFETs during assembly.

MAPPING TO THE SUSTAINABLE DEVELOPMENT GOALS

Table 2 realtes the project with the United Nations Sustainable Development Goals [10].

Table 2. Mapping of the project to the UN Sustainable Development Goals.

Goal	**Relevant target**	**Connection to this project**
SDG 9: Industry, innovation and infrastructure (primary)	9.c (access to information and communication technology); 9.5 (research and technological capability)	Investigates a low-cost, passive way to improve indoor wireless coverage with commodity laminate and components (listed component cost about INR 2 835).
SDG 7: Affordable and clean energy (indirect)	7.3 (energy efficiency)	A reflecting surface has no RF amplifier; its control power is set by the diode bias (about 37 mW per ON element in this design, Section 6.3). Whether this yields a system-level energy saving over active repeaters has not been evaluated.
SDG 4: Quality education (indirect)	4.4 (relevant technical skills)	Combines full-wave simulation, system modelling, PCB design and measurement planning in one student project.

CHAPTER 1

INTRODUCTION

1.1 Indoor propagation at 2.4 GHz

Most indoor wireless traffic in homes, laboratories and hostels is carried by Wi-Fi in the 2.4 GHz band. Inside a building the signal reaching a receiver is the sum of a direct component, when one exists, and many components reflected, diffracted and scattered by walls, furniture and people. Two effects follow [11]. The walls and floors are responsible for the addition of penetration loss. The multipath components are also responsible for addition in phases that change over distances of the order of a wavelength, which is 122 mm at 2.45 GHz; moving a device by a few centimetres can change the received power by several decibels. The first effect produces coverage gaps; the second makes any single measurement of signal strength a poor description of a location. The usual remedies are to add radiating hardware: more access points, mesh nodes, repeaters or active relays but the downside to this is that they all come with thier own RF chain, power supply, cabling and provides its own contribution to co-channel interference. A reconfigurable intelligent surface (RIS) approaches this the problem differently. Instead of adding transmitters or regenrating signal it dwells o it changes how an existing surface in the room reflects the signal that is already present.

1.2 Reconfigurable intelligent surfaces

An RIS is a planar array of sub-wavelength elements whose reflection coefficient can be switched electronically [12]. Each element re-radiates the incident field with a phase (and, in practice, an amplitude) that depends on its state. If the states are chosen so that the re-radiated fields arrive at a receiver in phase with each other and with the direct path, the received power increases; other choices move the reflected energy elsewhere. Because the elements only reflect, an RIS has no power amplifier and no receive chain; its power consumption is set by the control electronics and the switching devices.The simplest practical element has two states. In the coding-metasurface description of Cui *et al.* [13], the two states are labelled 0 and 1 and ideally differ in reflection phase by 180°; a spatial pattern of 0s and 1s then acts like a reconfigurable phase grating. A PIN diode is a convenient switch for this purpose at microwave frequencies: forward-biased it behaves as a small resistance, and unbiased or reverse-biased it behaves as a small capacitance, so switching the diode changes the electrical length and resonance of the element [14].

From the receiver's point of view, the effect of the surface can be written as

$$E_{\mathrm{RX}} = E_{\mathrm{direct}} + \sum_{n=1}^{M} \Gamma_n h_n g_n, \tag{1.1}$$

where Γ_n is the reflection coefficient of element n, h_n describes propagation from the

transmitter to the element, and g_n describes propagation from the element to the receiver.

1.3 Limitations of current approaches

Active solutions (repeaters, relays and additional access points) are effective but consume power continuously, need mains supply or cabling, and add interference. A passive RIS avoids these costs but has its own limitations, several of which recur in this project:

- **Multiplicative path loss.** Because the reflected path is the product of two links, a surface with a small aperture produces a weak reflected component unless it is placed near one end of the link [15].
- **Phase quantisation.** A one-bit element cannot synthesise a continuous phase gradient. Even with ideal 180° states, the steered beam loses about 3.9 dB relative to continuous phase control [16], and periodic one-bit codes produce mirror-image beams [13].
- **Non-ideal elements.** A practical element rarely reaches 180° separation, and its reflection amplitude varies with its state because resistive loss peaks near resonance [17].
- **Bias and control hardware.** Every element needs a DC path to its switch. Bias lines couple to the radiating structure and must be isolated at RF; for larger arrays the routing and control lines become a significant part of the design [18, 19].
- **Validation.** Hardware-aware design and measurement-based characterisation in realistic environments remain open needs; simulated gains do not transfer automatically to a cluttered room [20, 21].

1.4 Problem statement

Indoor wireless links suffer from attenuation, multipath fading and coverage gaps, and the conventional fixes add active RF hardware that relies on signal regeneration and high-power systems. The problem addressed in this project is to design, simulate and prototype a low-cost, PIN-diode-controlled reconfigurable surface operating near 2.45 GHz, and validate and observe the design performance through simulation and a controlled experiment to produce a measurable and repeatable change in a Wi-Fi link.

1.5 Objectives

This project investigates a low-cost, PIN-diode-controlled reconfigurable intelligent surface (RIS) operating near 2.45 GHz. The work combines electromagnetic simulation, system-level modelling, PCB design and preparation for experimental validation. The specific objectives are:

- Design a PIN-diode-switched unit cell for 2.45 GHz and characterise its reflection magnitude and phase in both diode states.

- Quantify how DC bias lines disturb the unit cell and select an RF choke that restores its phase response.
- Study 2×2 and 4×4 arrays of the cell, one-bit phase coding, beam steering and the effect of aperture size.
- Estimate, with a system-level model, how transmitter-surface and surface-receiver distances and the array size affect the received signal.
- Design a PCB for a single unit cell and for a 4×4 array with individually controlled elements, and to procure the components.
- Design and test in a controlled experiment in which two ESP32 boards measure the effect of the unit cell at varying distance of Wi-Fi link while an Arduino switches the diode.

1.6 Contributions

The principal technical contributions are:

- Design and simulation of a reconfigurable intelligent surface (RIS) unit cell operating at 2.45 GHz, using PIN-diode switching to investigate the reflection-phase response diffrent states.
- Analysis of the unit-cell reflection characteristics, including phase difference and reflection magnitude, to evaluate binary phase control for wireless signal manipulation.
- Design, fabrication, and assembly of the RIS unit-cell PCB.
- Experimental evaluation of Wi-Fi signal strength using ESP32-based transmitter and receiver nodes under different indoor propagation conditions, comparing direct-path, obstructed-path, and RIS ON/OFF configurations using (RSSI) measurements.
- Assessment of the measured results against simulation expectations, identifying practical limitations associated with the single-unit-cell prototype.

CHAPTER 2

LITERATURE REVIEW AND DESIGN REQUIREMENTS

This chapter reviews the literature that bears directly on the design decisions of the project and converts it into a set of requirements (Section 2.7).

2.1 Programmable surfaces and the RIS concept

The concept of a surface whose scattering can be programmed element by element was realised by the coding metamaterials of Cui et al. [13]. Each element is in one of a small number of discrete states and the spatial sequence of these states (the 'code') dictates the scattered field. With two states separated by 180 degrees, periodic codes produce symmetric beams of reflection and pseudo-random codes produce diffuse reflection; the surface becomes a programmable device if the states are electronically switchable.

Communication research then treated such surfaces as part of the channel. Wu and Zhang [22] optimised the reflection phases of an intelligent reflecting surface jointly with the access point beamformer and showed that the received power can grow with the square of the number of elements. Di Renzo *et al.* [12] reviewed how these surfaces work physically and how they should be modelled, and the ETSI group reports define use cases, channel models and practical considerations [4–6]. More recent surveys describe the engineering design and full-wave analysis of reconfigurable surfaces [23], the hardware and deployment questions faced by industry [24], and the engineering challenges that remain before RIS can be deployed in 6G systems [20]. Liang and Liang [25] survey systems that use several surfaces together, which is the configuration implied by the "multi-surface" aim in the title of this project; they point to deployment and coordination complexity as the main obstacles.

2.2 Switching devices and one-bit phase control

Reconfigurable reflectarrays have used PIN diodes, varactors, MEMS switches and liquid crystals to tune their elements [14]. PIN diodes dominate low-cost designs below about 10 GHz: they switch quickly, need only a few milliamperes of forward current and are available in small, inexpensive packages. Their drawback is that they are binary in practice, which leads naturally to one-bit elements. Yang *et al.* [18] built a 1-bit, 10×10 reconfigurable reflectarray and showed experimentally that careful element and bias design is needed to obtain usable beam scanning from binary states. Dai *et al.* [26] built 256-element, two-bit RIS prototypes at 2.3 GHz and 28.5 GHz using PIN diodes, and Pei *et al.* [27] built an 1100-element RIS at 5.8 GHz and evaluated it in indoor and outdoor field trials. The cost of discrete phases is well understood. With b-bit phase control, Wu and Zhang [16] showed that the squared power gain with the number of elements is preserved but

a constant loss is incurred; for one bit this loss is $(2/\pi)^2$, about 3.9 dB. Periodic one-bit codes also produce a mirror-image beam, because a real-valued ± 1 pattern cannot favour one side of the specular direction over the other [13]. These two effects are properties of one-bit coding even with ideal 180° elements; a practical element adds further loss when its phase separation falls short of 180° or its two states reflect with different amplitudes. The power consumed by the switches is not negligible when arrays grow. Chen [28] included PIN-diode power in the energy-efficiency analysis of RIS-assisted systems, and Chen *et al.* [29] demonstrated a low-power two-dimensional reflecting surface with high-speed serial control. Both bear on how far the bias scheme of this project can scale.

2.3 Element losses and amplitude–phase coupling

A resonant element can only produce a large change in reflection phase near resonance and that's precisely where the resistive and dielectric losses are also going to be highest. Abeywickrama *et al.* [17] model this coupling explicitly in that the reflection amplitude of a practical element will dip as its phase passes through resonance and beamforming that ignores this dip will perform worse than the theory would suggest. The implication for a one-bit design is that the state closer to resonance will reflect less and so the two states will differ in both amplitude and phase. On an inexpensive FR4 laminate the dip is deeper than it is on a low loss microwave substrate. Shabanpour *et al.* [30] take this one stage further and argue that beamforming models should go beyond the simple diagonal, element by element description of the surface. For small arrays with strong mutual coupling (as in the arrays studied in Chapter 5) the element response measured in a periodic environment is only an approximation.

2.4 Aperture, array scaling and path loss

Tang *et al.* [15] derived and measured free-space path-loss models for RIS-assisted links. In the far field of the surface, the received power is proportional to the square of the number of elements and inversely proportional to the squares of both the transmitter–surface and surface–receiver distances; in the near field the dependence changes. This model underlies the distance and array-size study in Section 5.7. Deployment studies show how strongly placement matters indoors: Kilcioglu and Oestges [31] used ray tracing to optimise RIS size and position for indoor coverage, and Poyda *et al.* [32] addressed how to adjust wide beams quickly in practical deployments. Work on complexity reduction [33] and on RIS for integrated sensing and communication [34] shows the range of system problems that build on the same element models, but these are beyond the scope of a single-cell prototype.

For the array geometry itself, classical array theory applies [35]. An element pitch below one wavelength keeps only the specular Floquet mode propagating under normal incidence, and a pitch of about half a wavelength allows steering over a wide angular range without grating lobes. The half-power beamwidth narrows in proportion to the aperture length, so a 4×4 array at

half-wavelength pitch still has a beamwidth of roughly 26° (Section 5.4).

2.5 Bias networks, control and hardware implementation

Every switched element requires a DC path and every DC path is a conductor adjacent to a resonant structure. Hardware reviews [19, 20] point to bias routing, RF isolation, fabrication tolerance and control line count as the key issues. The typical solutions are RF chokes or high impedance lines in series with the bias, radial stubs and routing the control lines on a separate layer behind the ground plane [14, 18]. The latter option requires a multilayer board and increases cost which is why a two layer FR4 board was chosen for this project (Section 6.6). Lumped chokes have their own set of problems in that a surface mount inductor has parasitic capacitance and a self resonant frequency so its impedance at 2.45 GHz will not be anywhere near ωL [36].

2.6 Experimental validation

Indoor experiments with binary surfaces predate the RIS label. Kaina *et al.* [37] used a binary-tunable metasurface in a reverberating room in the 2.4 GHz Wi-Fi band and shaped the field at chosen locations by optimising element states, demonstrating that even binary control can change indoor wireless fields measurably. Field trials with larger PIN-diode surfaces have reported clear improvements in received power in favourable geometries [27, 38], and measurement campaigns at 6 GHz have characterised RIS-assisted MIMO channels outdoors [21]. A common thread is that effects are large only when the surface is large compared with a wavelength, when it is placed close to one end of the link, or when the direct path is weak; a small surface in a strong line-of-sight link produces a small change that must be separated from multipath variation by careful experimental design.

2.7 Design requirements

Table 2.1 turns the review into requirements for this project. Where a requirement is quantitative, the number follows from a calculation in this report rather than from a single publication.

Table 2.1. Design requirements derived from the literature and the project constraints.

Requirement	Target	Justification
Operating frequency	2.45 GHz, covering Wi-Fi channels 8 and 9 (2447 and 2452 MHz)	Experiment uses ESP32 Wi-Fi in 2412 MHz to 2484 MHz [1, 3]
Phase states	One bit; $\Delta\phi$ as close to 180° as possible and at least about 140°	Equal-amplitude penalty $20\log_{10}\sin(\Delta\phi/2)$ stays below 0.6 dB for $\Delta\phi \geq 140°$
Reflection magnitude	High and similar in both states	The switchable component scales with $\lvert\Gamma_{ON} - \Gamma_{OFF}\rvert$; amplitude imbalance costs gain [17]
Element pitch	$\leq \lambda_0$; about $\lambda_0/2$ (61 mm)	Single propagating Floquet mode; no grating lobes over wide steering angles [35]
Aperture	As large as the budget permits	Coherent gain $\propto M^2$; far-field path loss $\propto (d_1 d_2)^2$ [15, 22]
Bias network	DC path with high RF impedance; minimal change to Γ	Bias lines detune elements [18, 19]
Control	Independent switching of each element; low static power	One-bit coding needs per-element states [28, 29]
Fabrication	Two-layer FR4, hand-solderable SMD parts	Budget and in-house assembly
Validation	Measurable with commodity Wi-Fi RSSI under controlled conditions	Small surfaces produce small changes [27, 37]

2.8 Design trade-offs

Phase separation against loss. Pushing one state close to resonance increases the phase separation but also the absorption in that state. On FR4 the trade-off is sharper than on low-loss laminates. The unit cell of this project reached about 141° with one state strongly absorbing (Chapter 4); recovering the remaining 39° would be worth less than reducing the loss of that state.

Element size against pitch. A patch that resonates near 2.45 GHz on FR4 is about 30 mm long, so a cell with room for the patch, the switch and the bias network is close to $\lambda_0/2$. The cell cannot be shrunk much without changing the element concept, and the pitch cannot grow much beyond $\lambda_0/2$ without narrowing the grating-lobe-free steering range.

Aperture against cost and complexity. Each step from 2×2 to 4×4 to 8×8 multiplies the coherent gain by about 12 dB in the far-field model, but also multiplies the board area, the number of diodes, chokes and drivers, and the number of control lines. A 4×4 board at 62 mm pitch is 300 mm square; the laminate purchased for the prototype is 100 mm × 75 mm.

Routing layers against RF cleanliness. With two copper layers, one of which is the ground plane, all bias and control traces share the layer with the patches. A multilayer board would hide them behind the ground plane at higher cost.

Experimental feasibility. Wi-Fi RSSI is cheap to measure but coarse (integer dBm) and strongly affected by multipath. A single cell can only change a 1 m link slightly, so the experiment has to be designed around repeated, interleaved measurements rather than single readings.

2.9 Gap addressed by this project

The literature establishes the principles and demonstrates large surfaces in favourable conditions but it often documented creates a gap for a simulated element to a small, inexpensive prototype and its various working as how the bias network changes the element, how the purchased parts differ from their simulation models, what the design actually contains, and what effect size a first prototype can be expected to show. This report folows this path for a 2.45 GHz PIN-diode surface on FR4 and perform simulation, design, fabrication and measurement.

CHAPTER 3

SYSTEM ARCHITECTURE AND DESIGN METHODOLOGY

3.1 Design workflow

The design workflow covers the first stages of requirements and electromagnetic modeling to array analysis, PCB designing, and scheduled measurement for the project and is summarized by Figure 3.1.

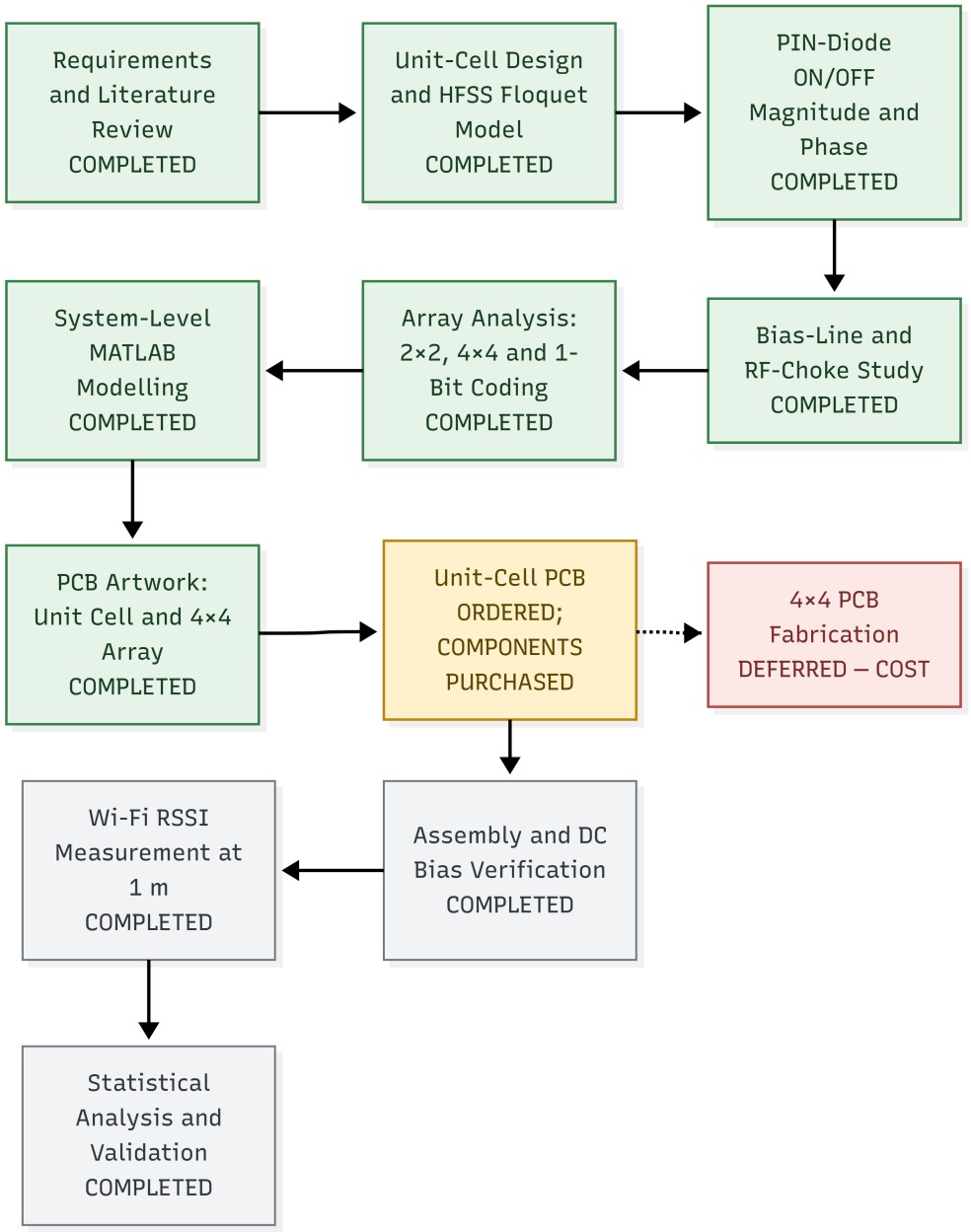


Figure 3.1. Block Diagram of the RIS Design and Validation Workflow.

3.2 Unit-cell geometry and materials

The unit cell design is a patch pair on a grounded FR4 substrate (Figure. 3.2). A rectangular main patch with two semicircular cut-outs on its lower edge and is separated by a narrow gap followed by a smaller secondary patch. The PIN diode is used to the bridge the gap at centre and when switching of diode is done it changes the effective resonant length of the element.The dimensions of unit cell design are mentione in the Table 3.1.

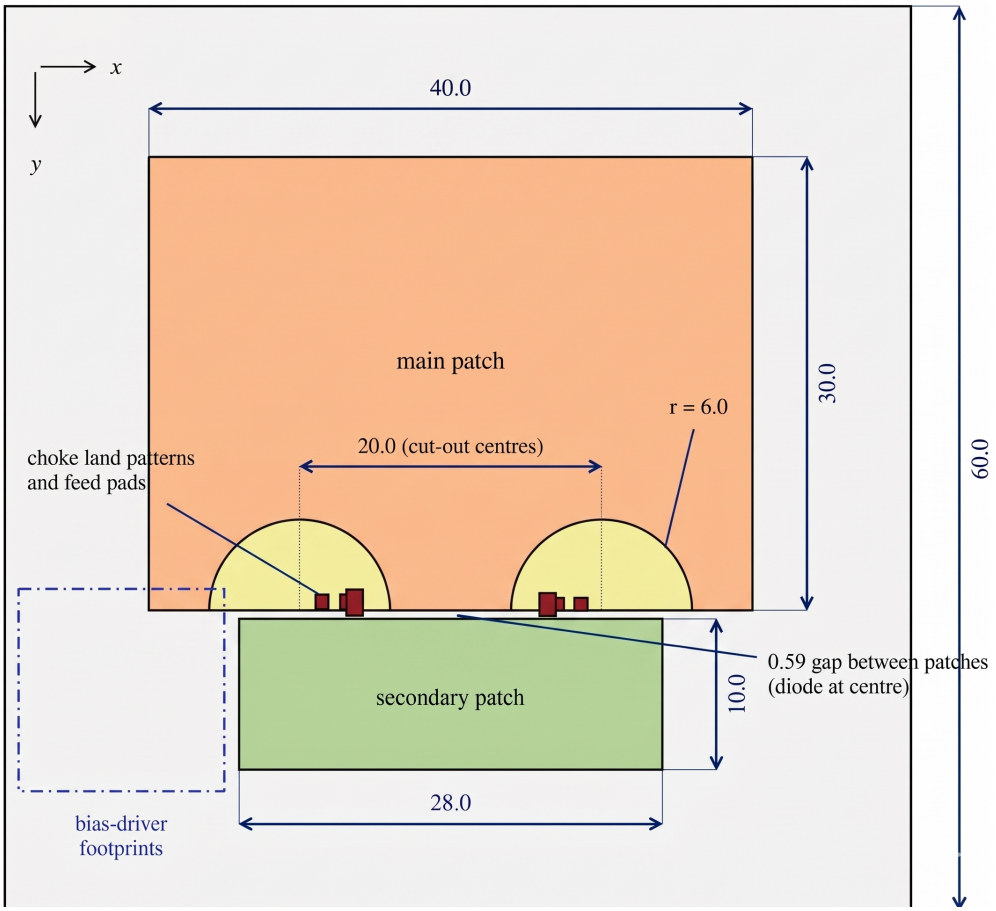


Figure 3.2. Unit-cell dimensions in millimetres.

Table 3.1. Unit-cell and array dimensions in mm and normalised to $\lambda_0 = 122.36\,\text{mm}$.

Feature	Size (mm)	Size (λ_0)
Unit-cell board outline (square)	60.0	0.490
4×4 element pitch (x and y)	62.0	0.507
Main patch, width × length	40.0×30.0	0.327×0.245
Semicircular cut-outs, radius; centre spacing	6.0; 20.0	0.049; 0.163
Secondary patch, width × length	28.0×10.0	0.229×0.082
Gap between main and secondary patch	0.59	0.005
Feed strips inside the gap, width	≈ 0.26	—
Rear-copper region (inferred ground plane)	59.0×59.0	0.482×0.482
4×4 board outline (square)	300.0	2.452

The unit-cell board size is (60.0 mm) with an array pitch of (62.0 mm calculated from frequency).

Substrate: The design uses FR4 as its substrate with a 1.5 mm thickness (Table 6.2). The permittivity, loss tangent and thickness used in HFSS are $\varepsilon_r = 4.4$ and $h = 1.5\,\text{mm}$.

3.3 Full-wave unit-cell simulation

The unit cell was simulated in Ansys HFSS with Floquet ports and periodic boundary conditions, which represent an infinite array of identical cells illuminated by a plane wave. With a period smaller than λ_0, only the fundamental Floquet mode propagates at normal incidence, and because the rear copper blocks transmission, any incident power that is not reflected is dissipated in the diode, the substrate and the copper. The reflection coefficient is read as the complex S_{11} of the Floquet port:

$$S_{11,\text{dB}} = 20\log_{10}|S_{11}|, \qquad \frac{P_{\text{refl}}}{P_{\text{inc}}} = |S_{11}|^2. \tag{3.1}$$

The second expression assumes power-normalised (generalised) scattering parameters and a single propagating Floquet mode. It describes the infinite periodic surface under the simulated incidence; it is not a measure of the efficiency of a finite surface or of a link.

The absolute phase of S_{11} depends on where the port reference plane is placed, but the reference plane is the same for both diode states, so the phase *difference* does not. The effective phase separation is computed on the circle:

$$\Delta\phi = \left|\,\text{wrap}_{[-180^\circ,\,180^\circ)}(\phi_{\text{ON}} - \phi_{\text{OFF}})\right|, \qquad \text{wrap}(x) = \big((x + 180^\circ) \bmod 360^\circ\big) - 180^\circ. \tag{3.2}$$

Figure 3.3 summarises the simulation methodology and shows the steps carried out through the entire simulation process. The MATLAB studies used models that were modeled after the HFSS analysis.

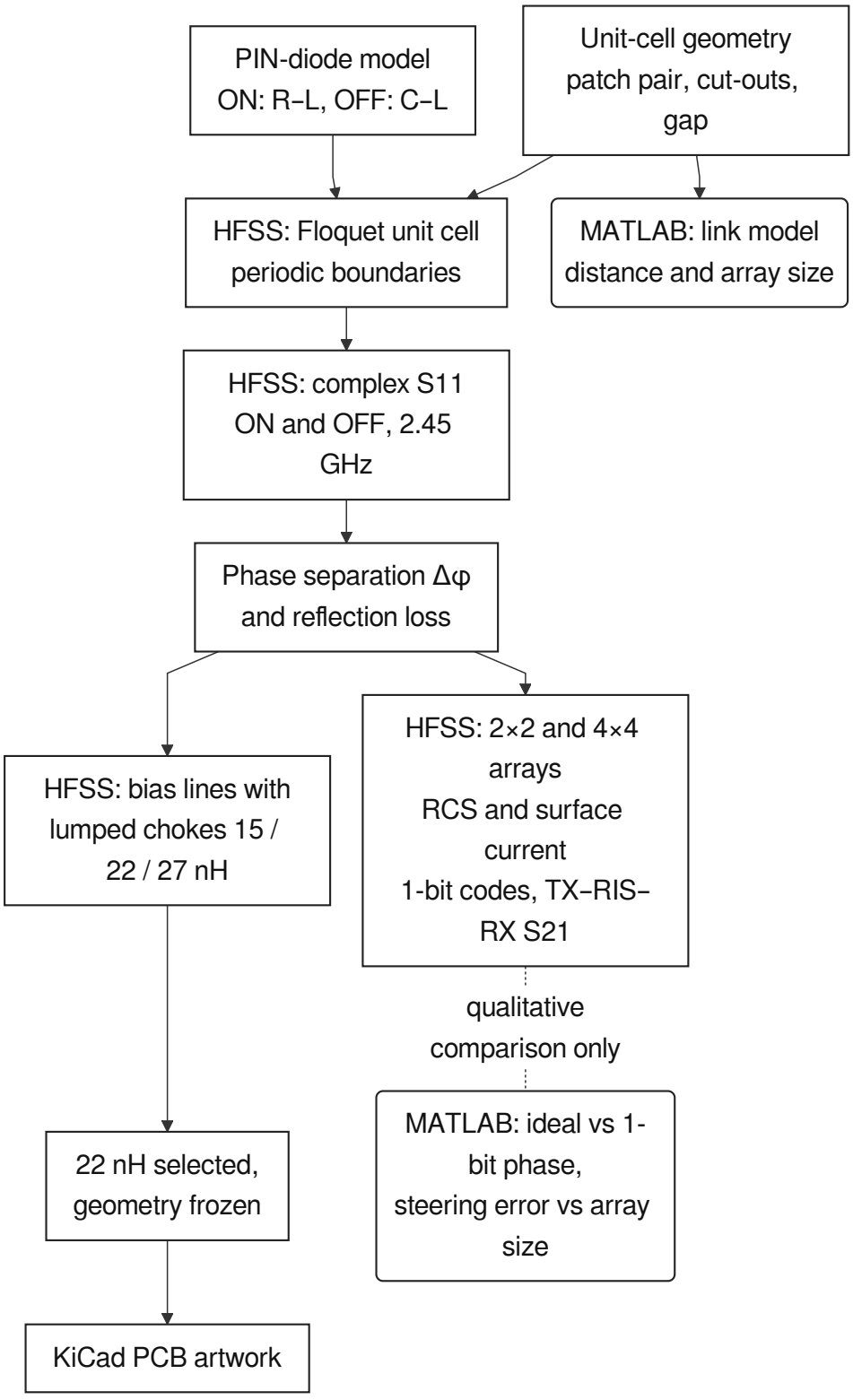


Figure 3.3. RIS simulation and design methodology.

3.4 PIN-diode modelling

At 2.45 GHz a PIN diode behaves as a linear element whose value depends on the DC bias as the signal period is far shorter than the carrier lifetime ($\tau = 75$ ns for the selected diode, giving a corner

frequency of $1/2\pi\tau \approx 2.1$ MHz). The usual equivalent circuits are a series resistance r_f with the package inductance L_S in the ON state, and a junction capacitance C_T with L_S (and a large parallel resistance) in the OFF state. Table 3.2 lists the datasheet values of the purchased BAR63-03W.

Table 3.2. PIN-diode parameters: datasheet values of the purchased BAR63-03W [39] and values used in HFSS.

Parameter	Datasheet value	HFSS model
Forward resistance r_f	1 Ω at $I_F = 10$ mA; 1.2 Ω at 5 mA (100 MHz)	2 Ω
Capacitance C_T	0.21 pF at $V_R = 5$ V, 1 MHz; 0.30 pF at $V_R = 0$ V, 0.1 GHz to 1.8 GHz	0 pF at ON State and 0.25 pF at OFF State
Package inductance L_S	1.8 nH (SOD323)	1.6 nH
Ratings	$V_R \leq 50$ V, $I_F \leq 100$ mA	—

The distinction between the two capacitance values matters for this design. The bias circuit described in Section 6.3 switches the diode between forward bias and zero bias; it never reverse-biases it. In the OFF state the relevant capacitance is therefore the zero-bias value of about 0.30 pF and the corresponding reactance at 2.45 GHz is $-217\,\Omega$.

3.5 Bias-network modelling

The switching of the diode is a crucial step in the design and requires that the main and secondary patches each be connected to a DC source. In the bias study, we routed two narrow copper traces from the cell edge to the two patches and added them to the HFSS model, first without isolation and then with a lumped inductor in series with each trace. The inductors were modelled as lumped RLC boundaries, and three standard values (15, 22 and 27 nH) were compared at 2.45 GHz by their ON and OFF phases, phase separation and reflection coefficient $|S_{11}|$.

3.6 Array modelling and state assignment

A finite 2×2 and 4×4 array of the idealised cell were simulated in HFSS with the diode state of each element set individually. Two kinds of study were made:

- **Scattering studies**, in which the array is illuminated by a plane wave and the bistatic radar cross section (RCS) and surface currents are computed. These were used for uniform states (all ON, all OFF) and for one-bit codes intended to steer the reflected beam to 10° and 20°.
- **A link study**, in which a transmitting dipole, the 4×4 surface and a receiving dipole are placed in a radiation-bounded region and S_{21} between the dipoles is computed for several codes: 1111, 0000, 1101, 1100, 1110 and 0101.

3.7 PCB Designing

After the bias study, the geometry and the 22 nH choke were frozen and the layout was drawn in KiCad, first for a single cell with its bias driver and then for a 4×4 array with one driver per element. The routing was done and the gerber files were genrated for unit cell manufaturing.

3.8 Experiment Setup

The experimental setup consists of two ESP32-WROOM-32 boards configured as a Wi-Fi transmitter and receiver, a fabricated RIS unit-cell PCB, and an Arduino to control the PIN diode through the bias circuit. The receiver records RSSI values and received-packet counts, which are subsequently processed in MATLAB. Measurements are performed under four conditions:

- An unobstructed link without the RIS.
- An obstructed link without the RIS.
- An obstructed link with the RIS in the OFF state.
- An obstructed link with the RIS in the ON state.

Tests are conducted at transmitter–receiver distances of 60 cm, 1 m, and 3 m. Samples with fewer than one received packet are excluded from the analysis. The mean RSSI for each condition is calculated and compared with the corresponding obstructed/no-RIS measurement to evaluate changes in received signal strength. The Arduino controls the diode state independently of the Wi-Fi communication link.

CHAPTER 4

UNIT-CELL DESIGN AND ELECTROMAGNETIC RESULTS

This chapter presents the simulated response of the unit cell, calculation, and examines the bias network and RF choke. All results are obtained in HFSS Floquet simulations of an infinite periodic surface (Section 3.3).

4.1 The HFSS unit-cell model

Figure 4.1 shows the HFSS model of the unit cell: the patch pair with its two semicircular cut-outs, the central gap holding the PIN diode, the grounded substrate and the Floquet port above the cell. In this idealised model the diode is represented by a lumped element whose value switches between the ON and OFF states.

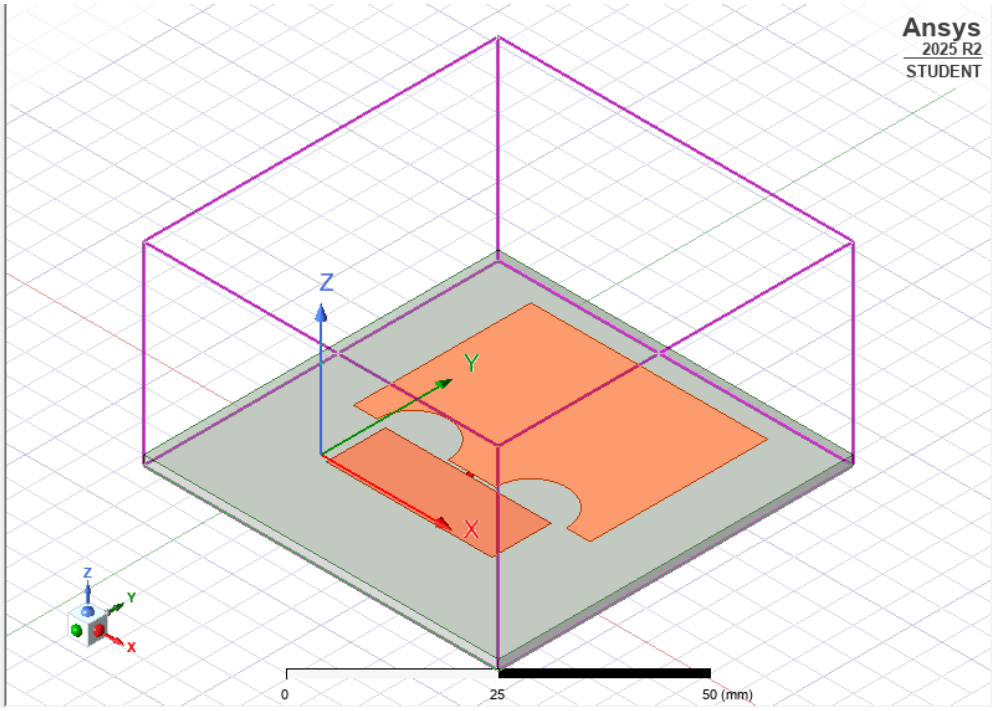
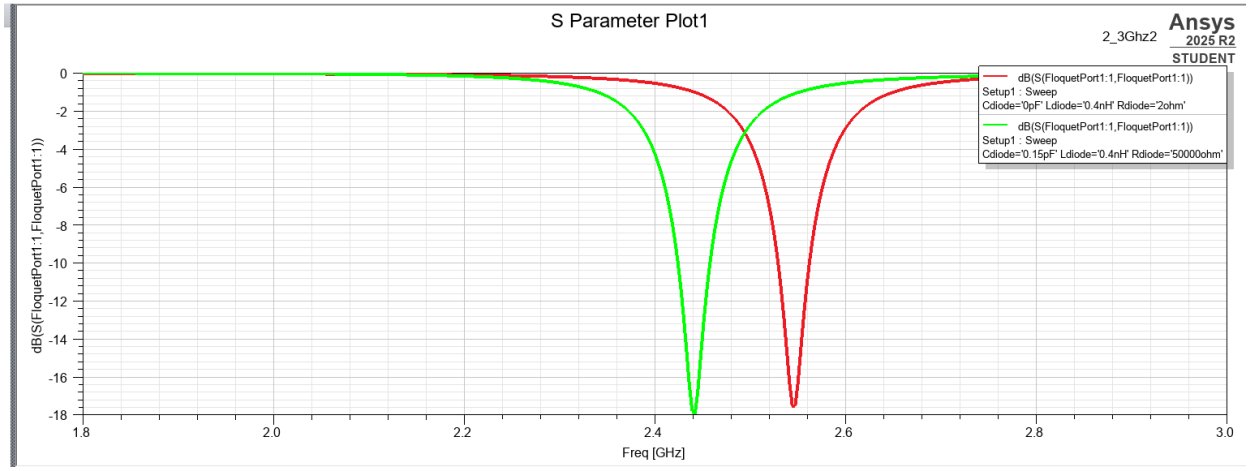


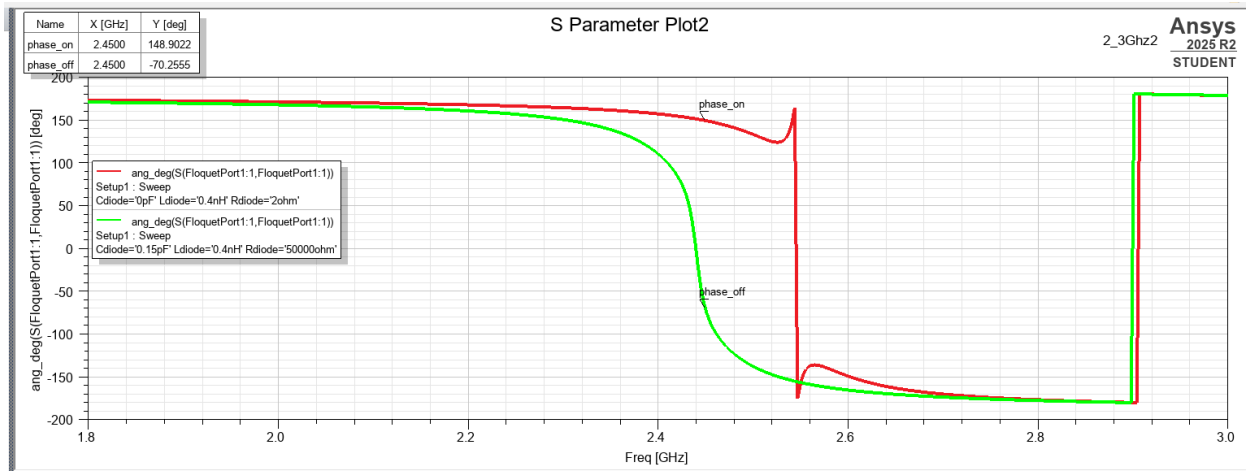
Figure 4.1. HFSS model of the unit cell with Floquet port and periodic boundaries .

4.2 Reflection magnitude

Figure 4.2(a) shows $|S_{11}|$ against frequency for both diode states. The ON state reflects strongly near 2.45 GHz with $|S_{11}| \approx -1$ dB, while the OFF state shows a resonant dip to about -10 dB.



(a) $|S_{11}|$ (dB), ON and OFF



(b) $\angle S_{11}$ (degrees), ON and OFF

Figure 4.2. Simulated reflection coefficient and phase of the unit cell.

The rear copper blocks transmission and the power that is not reflected is dissipated in the cell. The magnitudes and reflection calculated is given in Table 4.1.

Table 4.1. Reflected and dissipated power fractions from the simulated $|S_{11}|$ at 2.45 GHz.

Case (state)	$\lvert S_{11}\rvert$ **(dB)**	$\lvert S_{11}\rvert$	**Reflected**	**Dissipated**
Idealised cell, ON	≈ -1.0	0.891	79.4 %	20.6 %
Idealised cell, OFF	≈ -10.0	0.316	10.0 %	90.0 %
Bias lines, no chokes, OFF	-3.49	0.669	44.8 %	55.2 %
15 nH chokes, OFF	-9.22	0.346	12.0 %	88.0 %
22 nH chokes, OFF	-7.50	0.422	17.8 %	82.2 %
27 nH chokes, OFF†	-1.46	0.845	71.4 %	28.6 %

The OFF state of the unit cell dissipates about nine-tenths of the incident power. This is a direct consequence of operating that state near resonance on a lossy substrate. With a ground plane behind the cell, transmission is excluded and the resonant dip corresponds to absorption.

4.3 Reflection phase and effective phase separation

The phase plot (Figure. 4.2(b)) gives $\phi_{\mathrm{ON}} = 148.9^\circ$ and $\phi_{\mathrm{OFF}} = -70.3^\circ$ at 2.45 GHz. Applying (3.2),

$$\begin{aligned} \phi_{\mathrm{ON}} - \phi_{\mathrm{OFF}} &= 148.9^\circ - (-70.3^\circ) = 219.2^\circ, \\ \mathrm{wrap}(219.2^\circ) &= 219.2^\circ - 360^\circ = -140.8^\circ, \\ \Delta\phi &= |-140.8^\circ| = 140.8^\circ. \end{aligned} \tag{4.1}$$

The wrapped phase difference independently evaluates to 140.8°, obtained by subtracting the raw difference from 360°. The separation falls 39.2° short of the ideal 180° for one-bit coding. The absolute phases depend on the port reference plane, but $\Delta\phi$ does not, so this number characterises the element.

4.4 Equivalent-circuit checks on the diode

Table 4.2 converts the datasheet parameters of the BAR63-03W into reactances at 2.45 GHz. The package inductance resonates with the junction capacitance well above the band (6.9 GHz to 8.2 GHz), so the OFF state is capacitive, as intended.

Table 4.2. BAR63-03W impedances at 2.45 GHz from the datasheet values ($\omega = 2\pi f_0$).

State and condition	X_C (Ω)	$X_C + \omega L_S$ (Ω)	L_S–C_T **resonance**
OFF, $C_T = 0.21$ pF ($V_R = 5$ V)	−309.3	−281.6	8.19 GHz
OFF, $C_T = 0.30$ pF ($V_R = 0$ V)	−216.5	−188.8	6.85 GHz
ON, $r_f = 1\,\Omega$, $L_S = 1.8$ nH	—	+27.7 (with 1 Ω series)	—

The OFF-state reactance changes by about a third between the two capacitance values. Since the bias circuit leaves the diode at zero bias in the OFF state (Section 6.3), the 0.21 pF value is the appropriate one.

4.5 Bias lines and RF chokes

4.5.1 Effect of unisolated bias lines

Adding two plain copper traces from the cell edge to the patches significantly changed the cell response (Table 4.3). The phase separation decreased from 140.8° to 104.7°, while the OFF-state reflection magnitude increased from approximately −10 dB to −3.49 dB. At 2.45 GHz, these traces are electrically significant and become part of the resonant structure. Because they extend from the cell edges, periodic boundary conditions also extend them into neighbouring cells, effectively modelling bias lines across an infinite array. Although we initially described this change as a

collapse in $|S_{11}|$ due to RF leakage, the numerical results show increased reflection: the OFF state reflects about 45% of the incident power, compared with 10% before the traces were added. This suggests that the bias lines shifted the OFF-state resonance, reducing the absorption dip and the phase separation between the two diode states. These results highlight the need for effective RF isolation of the bias network.

4.5.2 Choke values

Table 4.3. Bias-line and RF-choke study at 2.45 GHz

Configuration	ϕ_{ON}	ϕ_{OFF}	$\Delta\phi$ **reported**	$\Delta\phi$ **recalc.**	$\lvert S_{11,\mathrm{OFF}}\rvert$
Idealised cell (no bias lines)	148.9°	−70.3°	140.8°	140.8°	≈ -10 dB
Bias lines, no chokes	139.8°	−115.5°	104.7°	104.7°	−3.49 dB
15 nH chokes	143.1°	65.8°	77.3°	77.3°	−9.22 dB
22 nH chokes (selected)	142.9°	−77.4°	139.7°	139.7°	−7.50 dB
27 nH chokes	142.9°	−88.5°	128.6°	128.6°	−1.46 dB†

All five configurations give rise to the same phase separations. The ON state phase is remains unchanged (ranging from 139.8 degrees to 148.9 degrees) which again points to the bias network having a dominant effect on the OFF state, whose resonance is within the operating band. The 15nH case is of particular interest in that the OFF state phase moves to +65.8 degrees, giving a phase separation of only 77.3 degrees. This implies that the bias path is resonating at around 2.45 GHz and the choke is therefore behaving as part of the resonant structure rather than as an RF isolator. With 22 nH the phase separation is back to 139.7 degrees (just 1.1 degrees below that of the idealised cell). That level of agreement was the basis for selecting the 22 nH choke.Figure 4.3 shows the HFSS model with bias lines and lumped chokes.

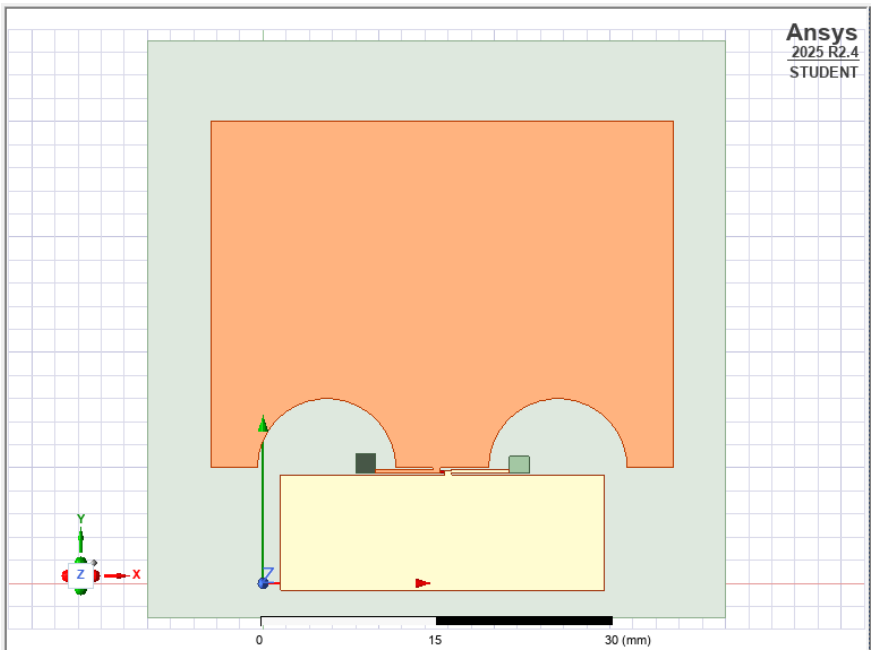


Figure 4.3. HFSS unit-cell model with bias lines and lumped RF chokes.

4.6 Summary of unit-cell findings

Table 4.4 collects what the unit-cell work establishes and how it feeds the next stages.

Table 4.4. Unit-cell findings, their limitations and the decisions they inform.

Finding	**Limitation**	**Consequence**
$\Delta\phi = 140.8°$ (idealised), $139.7°$ with 22 nH chokes	Floquet model; values read from plots	One-bit coding is feasible; phase error costs about 0.5 dB
OFF state dissipates 82–90 % of incident power	Magnitudes approximate	Main weakness of the cell; dominates the one-bit penalty (-4.8 dB for $\lvert b\rvert^2$)
Bias lines detune the OFF state strongly	HFSS traces end at the cell edge	RF isolation is mandatory; the choke is part of the resonator
Real choke $L_{\text{eff}} \approx 28$ nH to 49 nH at 2.45 GHz	Single-resonance estimates	Re-simulate with the vendor model before the array is built
Diode OFF capacitance is 0.30 pF at zero bias	HFSS value unknown	Check sensitivity of ϕ_{OFF} to C_T

CHAPTER 5

ARRAY ANALYSIS AND SYSTEM-LEVEL RESULTS

This chapter covers the finite-array simulations (HFSS), the one-bit steering studies (HFSS and MATLAB), the effect of array size, the full-wave link study and the MATLAB distance model.

5.1 Array-factor model

For a row of N identical elements with pitch d, illuminated at normal incidence, the field scattered in the direction θ (in the plane of the row) is the element pattern multiplied by the array factor [35]

$$\mathrm{AF}(\theta) = \sum_{n=0}^{N-1} w_n \, e^{\,jn\psi}, \qquad \psi = k_0 d \sin\theta, \qquad k_0 = \frac{2\pi}{\lambda_0}, \tag{5.1}$$

where $w_n = \Gamma_n$ is the reflection coefficient of element n. For a one-bit array,$w_n = a + b\,c_n$: the a term forms a uniform (specular) beam and the code $c_n = \pm 1$ forms the steerable part. A two-dimensional array with a code that varies only along x (columns) behaves, in the x–z plane, like (5.1) multiplied by the uniform factor of the rows.

With uniform weights, the half-power beamwidth narrows in inverse proportion to the aperture. Table 5.1 lists values computed numerically from (5.1) for $d = 62$ mm, the pitch of the 4×4 design. They describe the array factor alone, without element pattern or coupling.

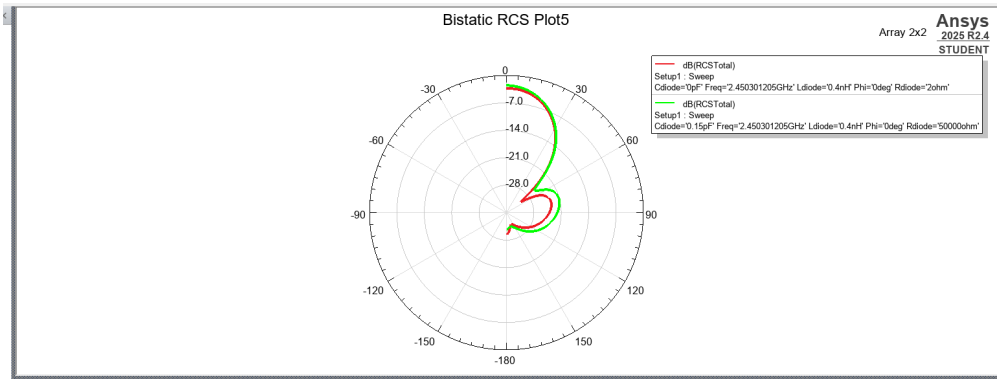
Table 5.1. Half-power beamwidth of the uniform array factor at broadside, $d = 62\,\mathrm{mm} = 0.507\lambda_0$.

Elements per side N	2	4	8	16	64
Aperture side (m)	0.124	0.248	0.496	0.992	3.968
HPBW (degrees)	59.1	26.0	12.6	6.3	1.6

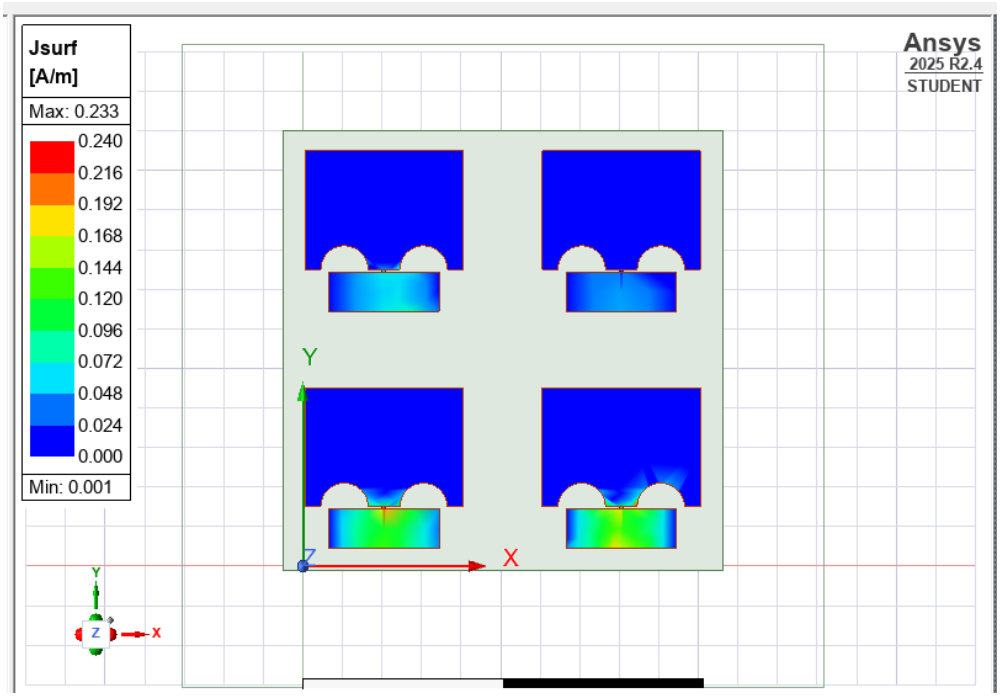
The 4×4 array therefore has a broadside beam about 26° wide. Steering results for this array must be read against that width: a change of a few degrees in the peak position of such a beam is a small change of the pattern.

5.2 2×2 Array study

The 2×2 array of the idealised cell was simulated under plane-wave illumination for uniform ON and OFF states (Figure. 5.1). Its bistatic RCS is broad and nearly omnidirectional in both states as the aperture is about one wavelength on a side (124 mm at 62 mm pitch), and Table 5.1 gives an array-factor beamwidth of about 59°, which is further broadened by the element pattern. A surface this small cannot form a directive beam; the 2×2 case is a baseline for the larger arrays.



(a) Bistatic RCS

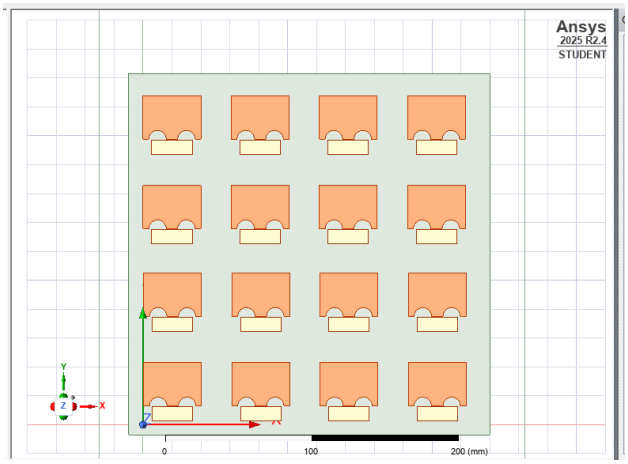


(b) Surface current density

Figure 5.1. 2×2 array of the idealised cell (HFSS; from the report).

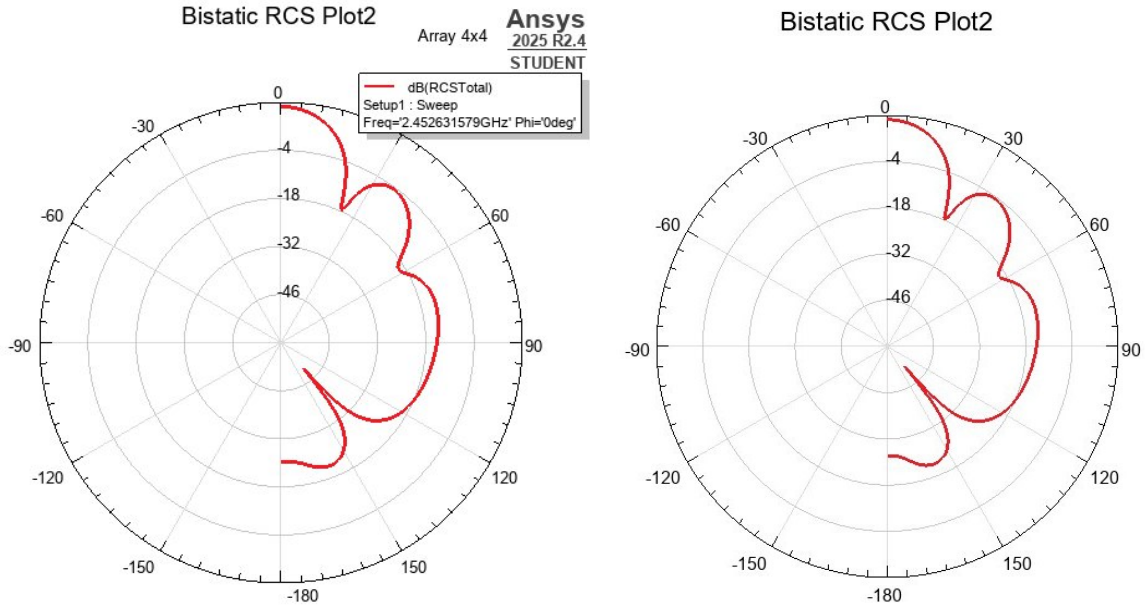
5.3 4×4 array: uniform states

The 4x4 array design (Figure. 5.2a) with all 16 elements in the same state the code is uniform, the phase gradient across the aperture is zero, and the surface behaves like a flat reflector. The simulation shows a dominant main lobe at broadside (0°) with clear nulls and side lobes. A broadside specular lobe also indicates that this simulation used normal incidence. The 4×4 aperture is four times larger in area than the 2×2, and its main lobe is correspondingly narrower (Table 5.1).



(a) 4x4 Array

In the array-factor approximation the two uniform states have the same pattern shape, because the weights are uniform in both cases; the OFF pattern should be lower by roughly $20\log_{10}(|\Gamma_{\mathrm{OFF}}|/|\Gamma_{\mathrm{ON}}|) \approx -9\,\mathrm{dB}$. A weak OFF lobe can look less directive on a linear or autoscaled plot. Differences in element pattern, edge effects and coupling between the two states could also change the shape. The two descriptions cannot be reconciled without the plots (Figure. 5.3) on a common scale.



(a) All elements OFF

(b) All elements ON

Figure 5.3. Bistatic RCS of the 4×4 array in the uniform states (HFSS; from the report).

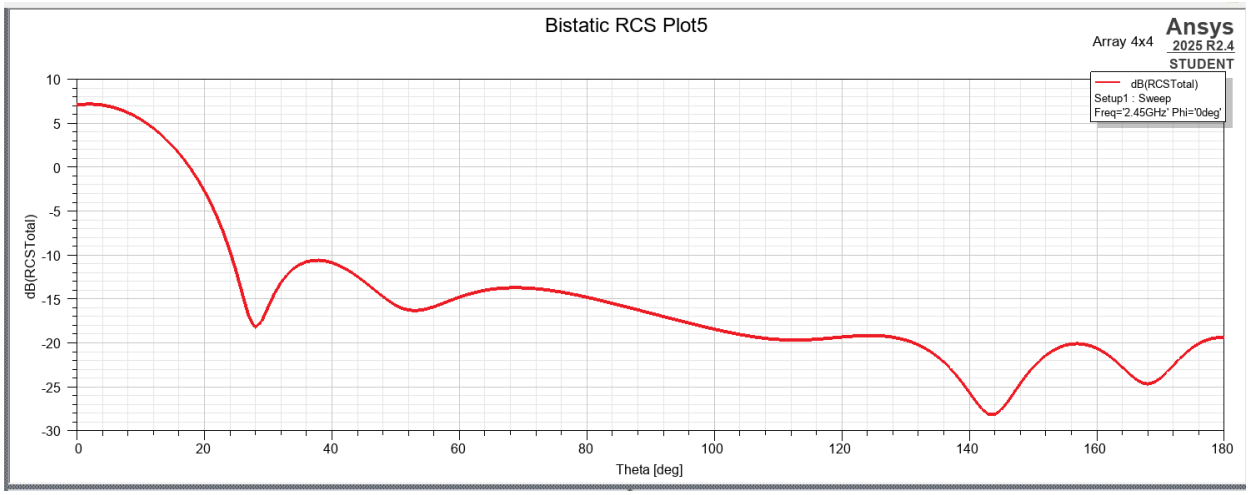
5.4 One-bit coding and beam steering

5.4.1 Results

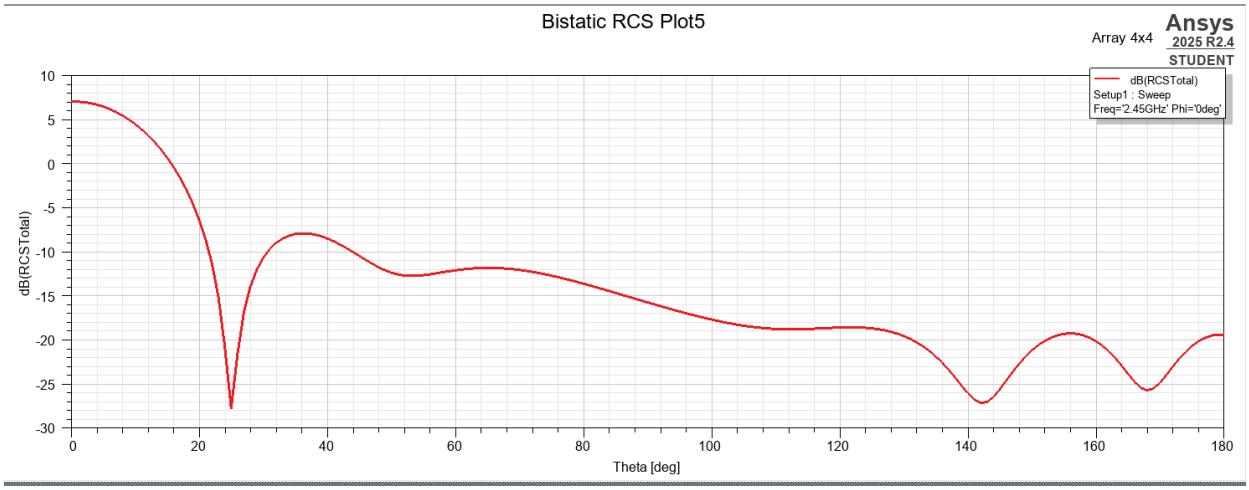
Two steering cases were simulated in HFSS on the 4×4 array (Figure. 5.4):

- a 10° target, using a quasi-random binary pattern, gave a main-beam peak at 10.0°;

- a 20° target, using a column-based binary gradient, gave a peak at 18.1°, a deviation of 1.9° that the report attributes to one-bit quantisation.



(a) 10° target



(b) 20° target

Figure 5.4. One-bit beam steering with the 4×4 array (HFSS; from the report).

5.4.2 Array-factor interpretation

A linear phase gradient that steers a normally incident wave to 20° advances the phase by 360° $(d/\lambda_0)\sin 20° = 62.4°$ per column at $d = 62$ mm, so the four columns would need 0°, 62.4°, 124.8° and 187.2°. Rounding each to the nearer of the two available states gives the code [1 1 0 0] (or its complement), which is also one of the codes in the link study of Section 5.6. The infinite periodic repetition of this code would steer to $\arcsin[\lambda_0/(4d)] = 29.6°$, because its period is fixed at four cells; on a finite 4-column aperture the peak is pulled inwards.

Figure 5.5 evaluates (5.1) for this case with isotropic elements and normal incidence. The code [1 1 0 0] produces two equal beams at ±22.8°, each 2.3 dB below the peak of the ideal continuous gradient; with a $\cos\theta$ element factor the peaks move to ±21.1°. The finite-aperture penalty is smaller than the asymptotic 3.9 dB because the beams of a four-element row are broad. Three conclusions follow.

- Under normal incidence the array factor places the steered beam *beyond* 20°, while HFSS gave 18.1°. Quantisation alone does not explain the HFSS value; the element pattern, coupling in the finite array, the incidence angle and the exact code all contribute. The simulation setup does not specify the incidence angle or the exact code mapping; these details should be recorded in any reproduction.
- A one-bit code under normal incidence produces a mirror beam of equal strength at the negative angle. The "parasitic quantisation lobe" near −30° reported in the MATLAB comparison is most probably this mirror beam; its position would equal the negative of the steered angle only for normal incidence, which again points to the need to record the incidence used.
- For the 10° case, the target lies inside the half-power width of the broadside beam (±13°). A quasi-random code can place the peak there, but the stated precision (10.0°) reflects the angular sampling of the plot more than the steering accuracy of a 26-degree-wide beam.

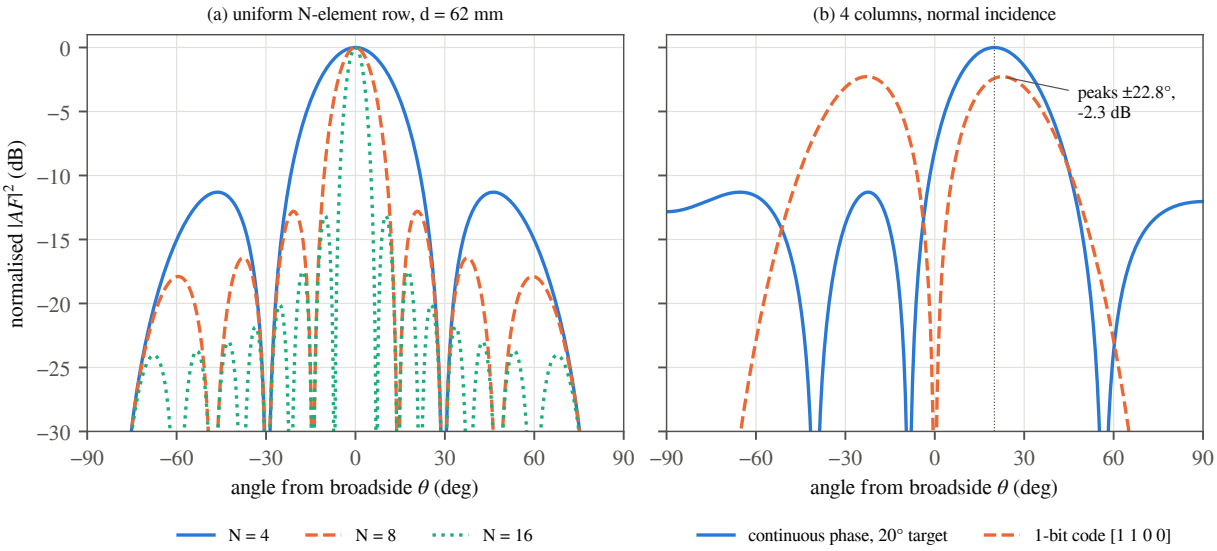


Figure 5.5. Analytical array factor: (a) uniform rows of N elements at 62 mm pitch; (b) four columns steered to 20° with continuous phase and with the one-bit code [1 1 0 0], normal incidence, isotropic elements.

5.5 Effect of array size

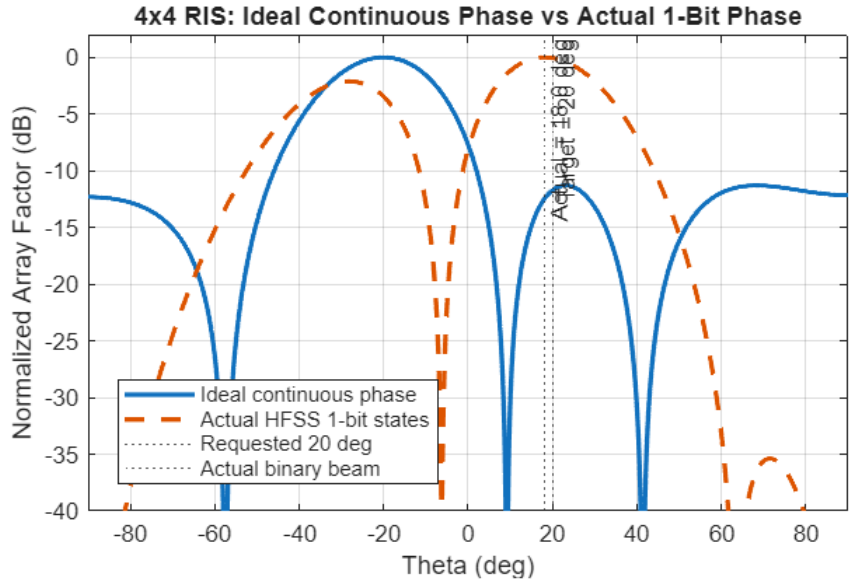
The simulation studies extended the one-bit analysis to larger apertures (Figure. 5.6). The simulation analysis yields three main results:

- One-bit pattern has a lower main lobe and higher quantisation lobes than the continuous-phase pattern;
- Steering error for the 20° target decreases rapidly as the array grows from $N = 4$ to $N = 64$ per side;
- Beamwidth contracts strongly from 4×4 to 64×64, with side lobes better separated from the main lobe.

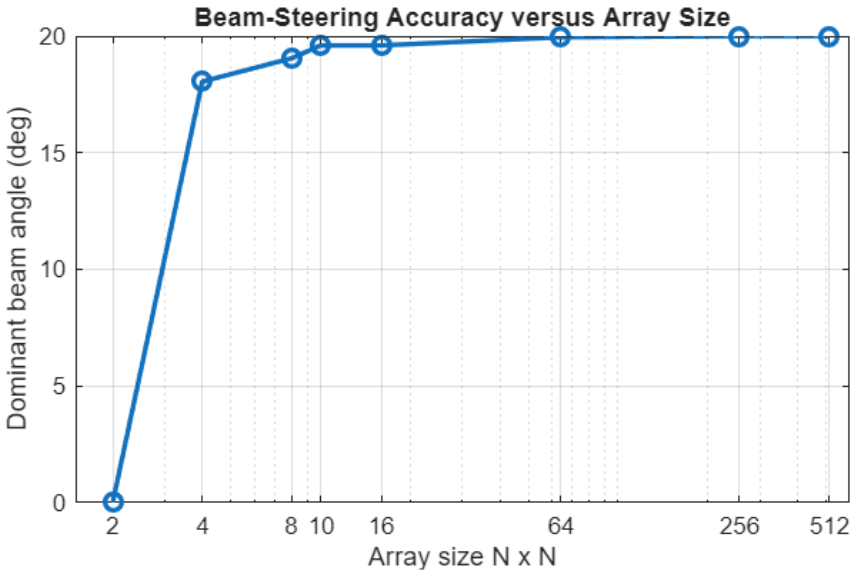
These trends agree with array theory: more columns allow the binary code to follow the ideal

gradient more closely, and a larger aperture narrows the beam (Table 5.1). They describe the coding geometry, not a practical design. A 64×64 array at 62 mm pitch would be about 4 m square with 4 096 diodes.

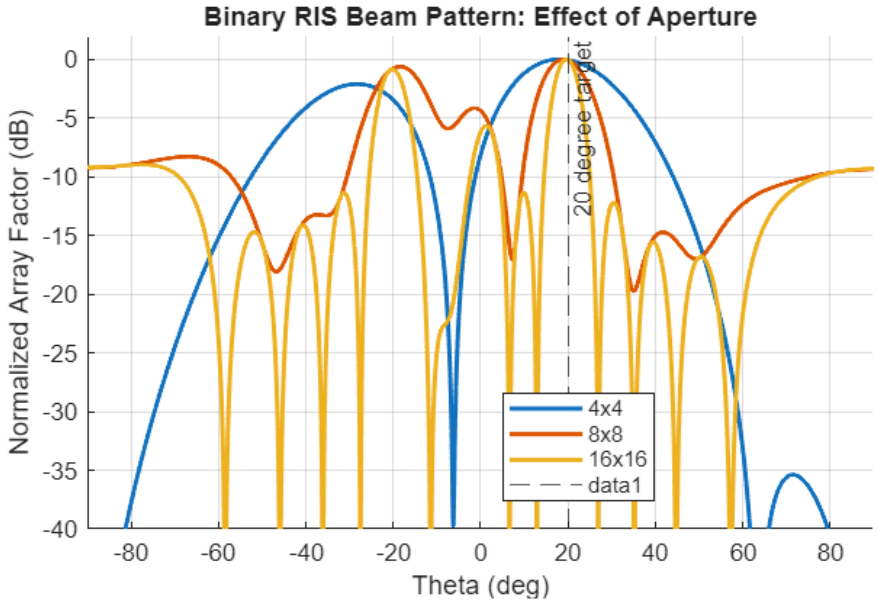
Increasing the element count from 16 to 64 increases the aperture area fourfold and, in the far-field model, the coherent reflected power by about 12 dB. It does not by itself guarantee a better link: the gain is realised only if the code is matched to the geometry, the element losses are acceptable, and the reflected path is significant compared with the direct path (Section 5.7).



(a) One-bit vs continuous



(b) Steering angle vs N



(c) 4×4 to 64×64

Figure 5.6. MATLAB array-factor studies of one-bit quantisation and aperture size .

5.6 Full-wave TX–RIS–RX study

The link study placed a transmitting dipole, the 4×4 surface and a receiving dipole in a radiation-bounded HFSS model and computed S_{21} between the dipoles for the codes in Table 5.2.

Table 5.2. Codes evaluated in the HFSS TX–RIS–RX study (1 = diode ON).

Code	Description
1111	All elements ON
0000	All elements OFF (reference)
1101, 1110	Mixed binary configurations
1100	Mixed binary configuration; one-bit approximation of a 20° gradient (Section 5.4)
0101	Alternating (checkerboard-type) distribution

The simulation indicates that the codes produce distinct S_{21} levels at 2.45 GHz, that 1100 gives the highest S_{21} among the mixed codes, and that the spread between codes is small at the simulated geometry (Figure. 5.7). The S_{21} values and the dipole positions are not recorded in the available material, so the size of the effect cannot be quoted. A small spread is what the analysis of Section 5.7 predicts whenever the direct dipole-to-dipole path is much stronger than the path via a 16-element surface. The study's main value is methodological: it established that the effect of the surface has to be evaluated for a specific TX–RIS–RX geometry, which motivated the distance study.

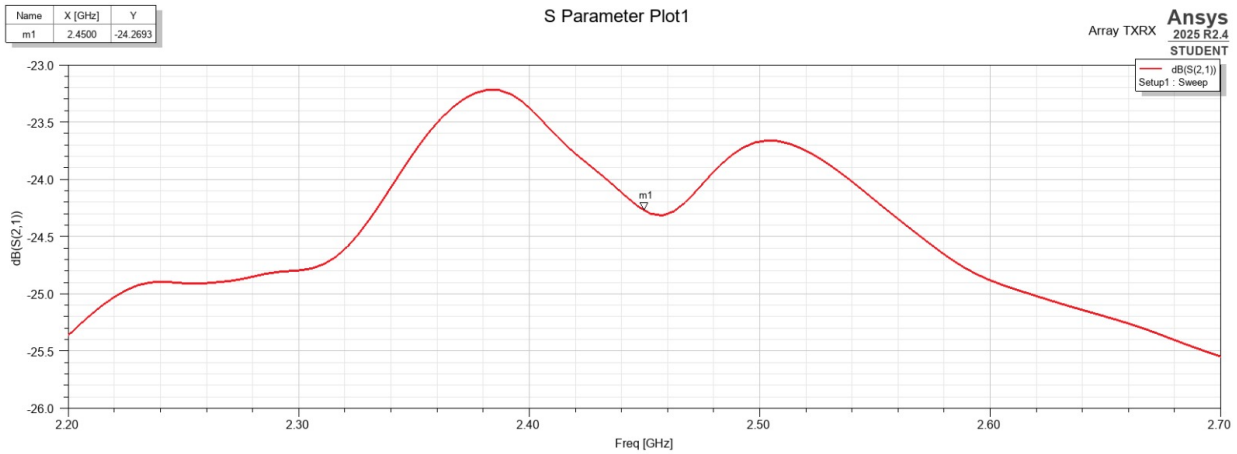


Figure 5.7. Simulated S_{21} between transmitting and receiving dipoles with the 4×4 surface in code 1100.

5.7 System-level distance and array-size study

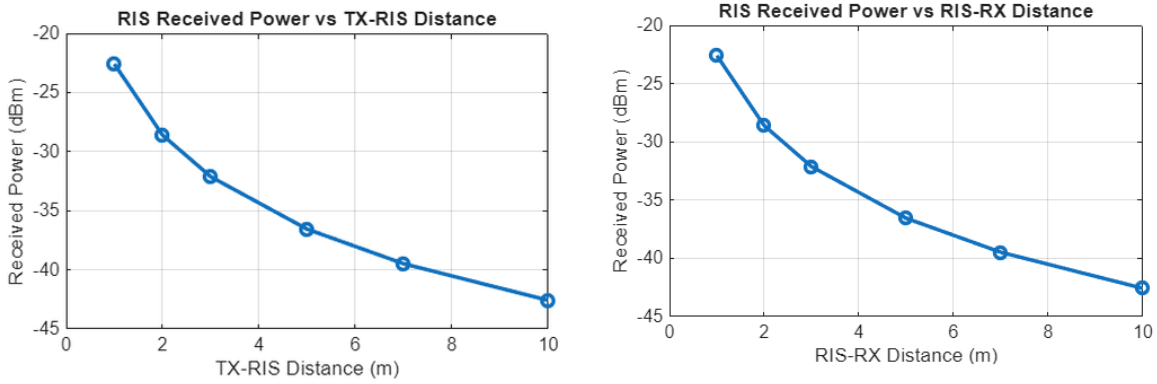
5.7.1 Results

The MATLAB link model gives the received power and a transmission response as functions of distance and array size (Table 5.3, Figure. 5.8).

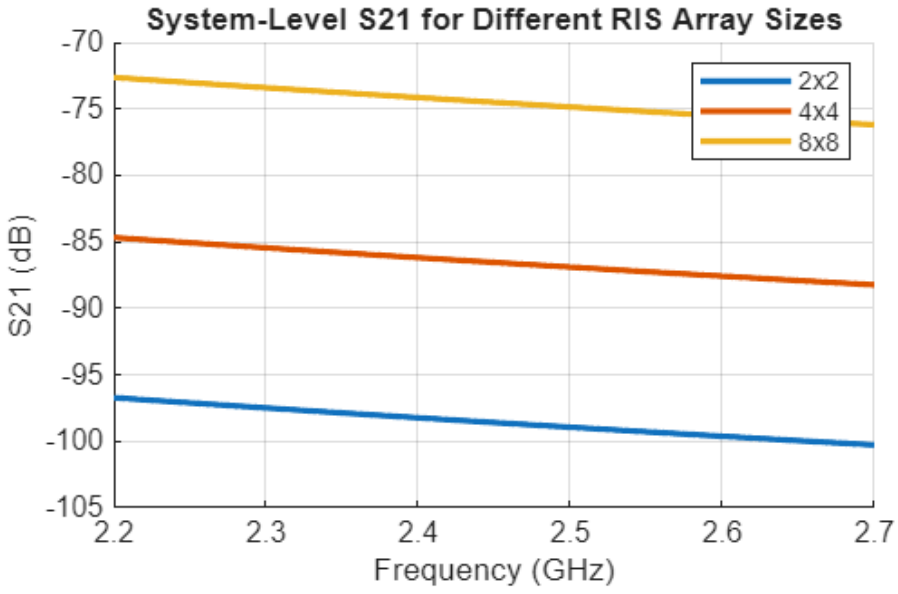
Table 5.3. MATLAB system-level results at 2.45 GHz.

Sweep	Reported result	Metric
TX–RIS distance 1 m to 10 m	−23 to −43	Received power
RIS–RX distance 1 m to 10 m	−23 to −42	Received power
Array size: 2×2, 4×4, 8×8	−97 dB, −87 dB, −74 dB	Reported array performance metric

0.5



(a) Received power vs TX–RIS and RIS–RX distance



(b) Response vs array size

Figure 5.8. MATLAB system-level study.

5.8 Limitations of the array and system results

- **Finite aperture.** A 4×4 array is about two wavelengths across. Its beams are wide (about 26°), steering is coarse, and edge effects and coupling are significant.
- **Element response.** The finite-array studies used the idealised cell. With the bias network, the OFF state changes (Chapter 4), and its high loss reduces the switchable part of the scattered field (−4.8 dB for the idealised magnitudes).
- **Propagation geometry.** The effect of the surface is small whenever the direct path dominates; the HFSS link study and the MATLAB model both show this.

CHAPTER 6

PCB DESIGN, COMPONENT SELECTION AND HARDWARE IMPLEMENTATION

This chapter documents the transition from the frozen simulation model to hardware: the unit-cell and 4×4 PCB design, the bias-driver circuit, the components bought, the fabrication constraints, and the proposed measurement hardware.

6.1 PCB Design Flow

After the choke study (Section 4.5) the cell geometry and the 22 nH choke were frozen and the layout was drawn in KiCad. Two layouts are designed: a single cell with its own bias driver and connection pads, and a 4×4 array in which every cell has a driver and an individual control line.

6.2 Unit-cell PCB

Figure 6.1 shows the unit-cell design. The board is 60.0 mm square. The patch pair occupies the centre: the 40 mm × 30 mm main patch with its two semicircular cut-outs, and the 28 mm × 10 mm secondary patch 0.6 mm below it.

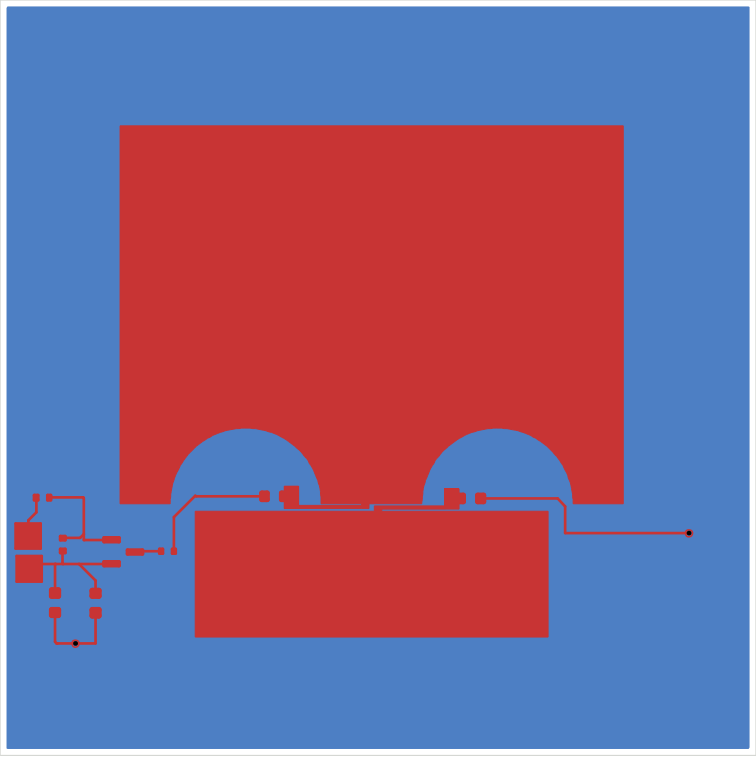


Figure 6.1. Unit-cell PCB design

Figure 6.2 shows the bias connections in more detail. Each semicircular cut-out has a two-pad footprint, with one pad connected to a rectangular feed pad. A narrow copper strip, approximately 0.26 mm wide, runs from each feed pad towards the centre of the gap between the patches. The left strip connects to the main patch, while the right strip connects to the secondary patch. The left footprint connects to the bias driver at the board's lower-left corner. A trace connects the right footprint to a via near the right edge, which links to the rear copper layer. These footprints provide locations for the RF chokes in the bias paths, while the PIN diode sits across the gap where the two strips meet.

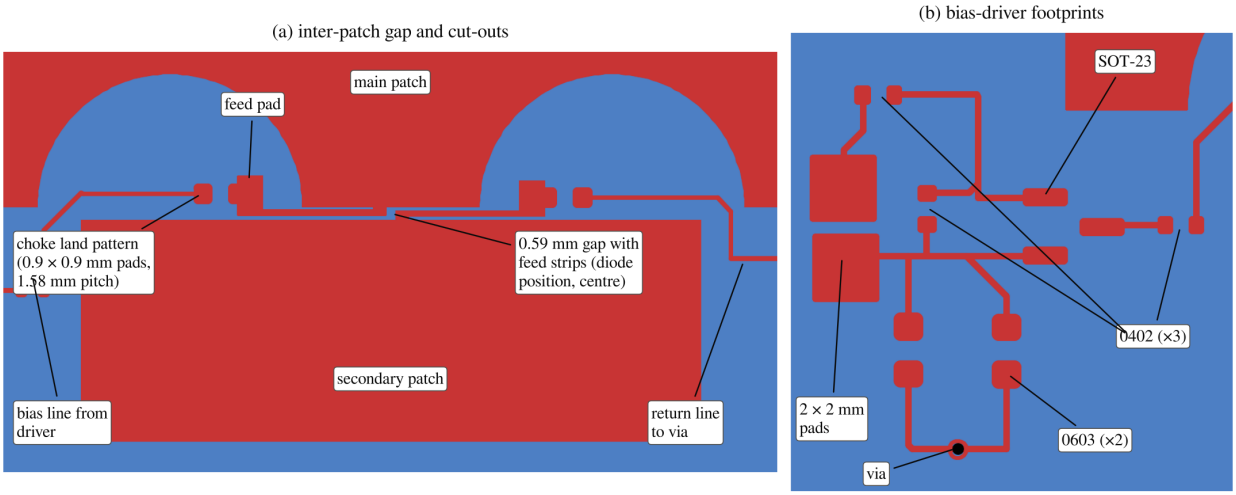


Figure 6.2. Close-ups of the unit-cell design: (a) inter-patch gap and cut-outs; (b) bias-driver footprints.

6.2.1 Footprint inventory

Table 6.1 lists every land pattern found on the unit-cell design, identified by its measured size and pitch, and compares it with the purchased parts.

Table 6.1. Land patterns on the unit-cell design compared with the purchased parts.

Land pattern (measured)	No.	Probable part	
SOT-23: pads 1.48×0.59 mm, pin pitch 1.90 mm	1	P-MOSFET	
0402: pads 0.55×0.64 mm, pitch 1.02 mm	3	Resistors	100 Ω, 220 Ω, 100 kΩ parts are 0402
0603: pads about 0.95×0.90 mm, pitch 1.57 mm	2	Decoupling capacitors	1 µF part is 0603; the 100 nF part is 0402
Two-pad pattern in each cut-out: pads 0.87×0.93 mm, pitch 1.58 mm, gap 0.70 mm	2	RF chokes	0402HP chokes are at most 1.09 mm long
Square pads 2.0×2.0 mm	2	Wire connections (control, 3.3 V)	—
Via pads, diameter 0.63 mm	2	Connections to rear copper	
SOD-323 land pattern	0	PIN diode	BAR63-03W is SOD-323

6.3 Bias-driver circuit

A 3.3 V bias circuit with an AO3401A P-MOSFET switch, 22 nH RF chokes and supply decoupling is used and tested.

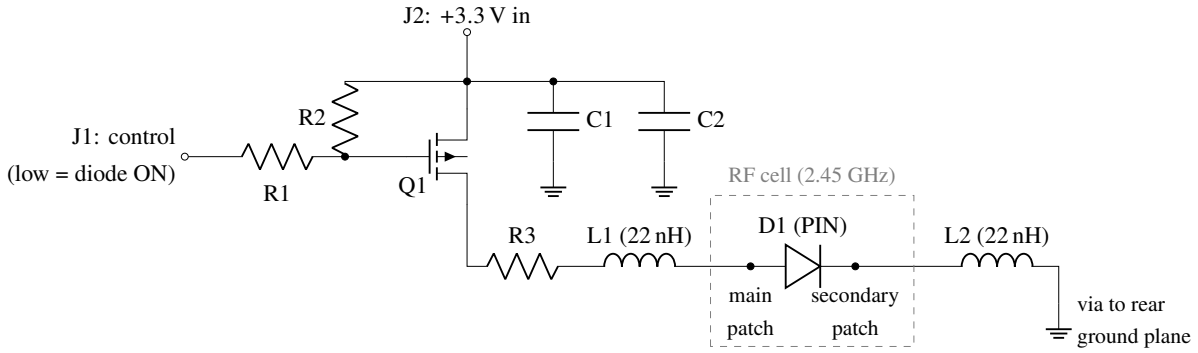


Figure 6.3. Bias-driver circuit inferred

Operation. Q1 is a high-side switch. When the control input J1 is pulled low, the gate is about 3.3 V below the source, the transistor conducts, and current flows through R3, the first choke, the main patch, the diode, the secondary patch and the second choke to the rear ground. When J1 is high, or left floating, the gate–source resistor R2 holds V_{GS} at zero and the diode is unbiased. The

logic is therefore active-low, and the surface defaults to the OFF state when the controller is reset or disconnected. R1 limits the current into the gate capacitance during switching.

Diode current. With the transistor on, the forward current is set by the series resistor:

$$I_F = \frac{V_S - V_F}{R_3 + 2R_{\mathrm{DC,L}} + R_{DS(\mathrm{on})}}. \tag{6.1}$$

The chokes contribute at most 0.16 Ω each [40] and the AO3401A at most 85 mΩ at $V_{GS} = -2.5$ V [41], so the denominator is essentially R_3. The datasheet gives the diode's forward voltage only at 100 mA (typically 0.95 V) [39]; at about 11 mA it is lower, so the range 0.75 V to 0.95 V is used. With $V_S = 3.3$ V:

$$R_3 = 220\,\Omega:\ I_F = 10.7\,\mathrm{mA}\text{ to }11.6\,\mathrm{mA}; \qquad R_3 = 100\,\Omega:\ I_F = 23.4\,\mathrm{mA}\text{ to }25.4\,\mathrm{mA}. \tag{6.2}$$

The bias-driver transient simulation gives approximately 11 mA to 12 mA for a 0 V to 3.3 V control signal, consistent with a 220 Ω series resistor. The value generally assumed for the ON state at this current the datasheet forward resistance is usually about 1 Ω, on the whole. Indeed, each ON element draws about 37 mW from the supply, so a 4×4 array with every element ON would draw about 0.59 W, at least for now. The resistor fades away 25 mW to 29 mW, within the 62.5 mW normally rating of the 0402 parts, so a 100 Ω resistor in that position would spread out 55 mW to 65 mW, at or beyond the rating.

Gate drive and controller logic levels. The AO3401A has $V_{GS(\mathrm{th})}$ between −0.5 and −1.3 V [41], so a gate pulled to 0 V with the source at 3.3 V switches it fully on. In the inferred circuit, a 3.3 V logic-high gives $V_{GS} = 0$ and a 5 V logic-high gives $V_{GS} = 1.7$ V; both turn the transistor off and both are within its ±12 V gate rating. The transient simulation used 3.3 V logic, matching the ESP32 controller planned at the initial design review. If the Arduino that now replaces it is a 5 V board, the circuit should still switch correctly, but this conclusion rests on the inferred schematic and must be confirmed.

OFF-state bias. The circuit never reverse-biases the diode. In the OFF state the diode sits at 0 V, where its capacitance is about 0.30 pF rather than 0.21 pF (Section 4.4).

Ground connection. The top copper has no ground pad; the ground is the rear copper, reached through the vias. The controller and the supply must share this ground, so a ground wire has to be attached to the rear copper or a ground pad added.

6.4 Proposed 4×4 PCB

Figure 6.4 shows the 4×4 design. The board is 300.0 mm square and carries 16 copies of the unit-cell copper on a 62.0 mm pitch in both directions, with the array centred on the board. Each cell has its own bias driver with the same footprints as the unit cell, except that the two connection pads are replaced by routed traces. A supply bus runs along one edge and feeds a supply line along each row. Each cell's control input has its own trace; the 16 control traces run along the rows to the opposite edge, where they end in two groups of eight at a spacing of about 2.5 mm, roughly the 2.54 mm pitch of a pin header, about 10 mm from the board edge. The board has 32 vias, two per cell, as on the unit cell.

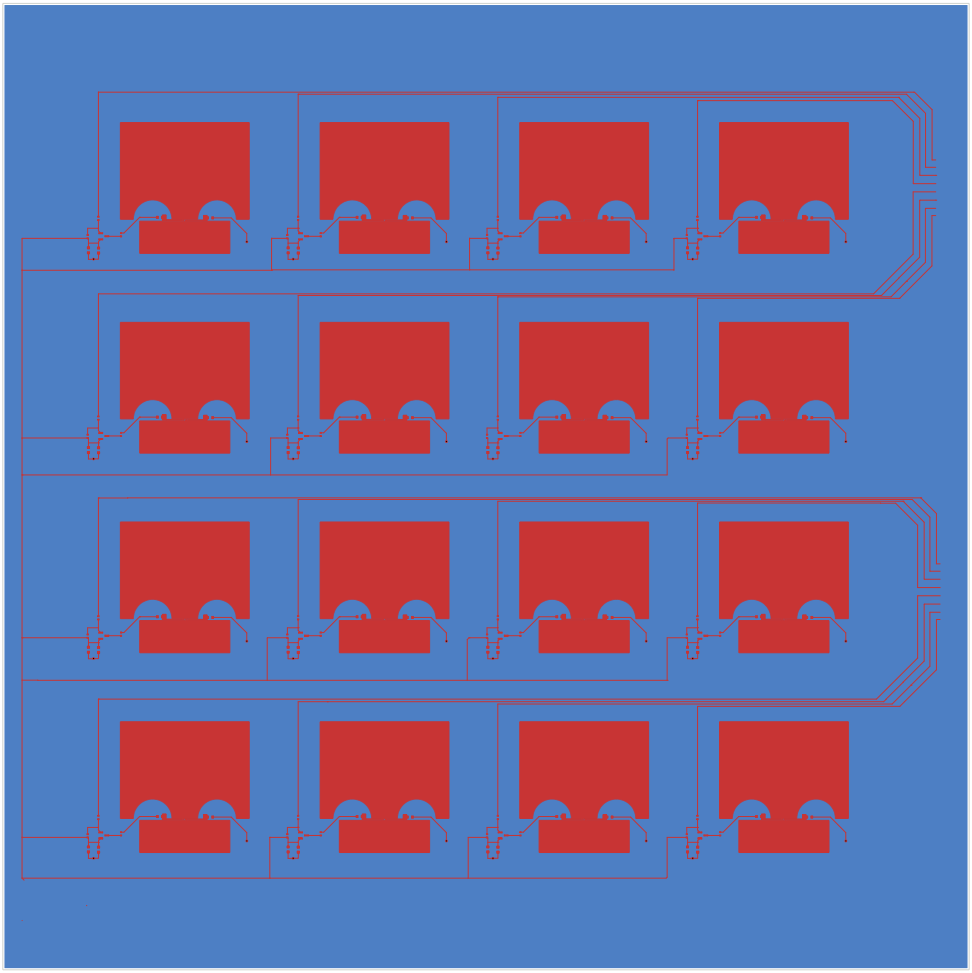


Figure 6.4. Proposed 4×4 PCB Design.

- **Pitch.** The element pitch of 62.0 mm is 2 mm.
- **Connectors.** The control traces and the supply bus are placed according to a connector land pattern.
- **Land patterns.** The choke and diode land-pattern issues of Section 6.2.1 are repeated in all 16 cells.

Table 6.2. Bill of materials

Component	Part / specification	Qty	Unit in Rs	Total in Rs	Intended use
PIN diode	Infineon BAR63-03W (BAR6303WE6327HTSA1); 0.21 pF, 1 Ω, 50 V, SOD-323	52	19.00	988.00	RF switch
RF choke	Coilcraft 0402HP-22NXGLU; 22 nH ±2 %, SRF 3.3 GHz, 0.16 Ω, 800 mA, 0402	37	11.00	407.00	Bias RF isolation
P-MOSFET	AO3401A (UMW); −30 V, 4.2 A, SOT-23	50	3.58	179.00	Bias switch
Resistor	Yageo RC0402FR-07220RL; 220 Ω, 1 %, 0402	50	0.38	19.00	Bias network
Resistor	Yageo RC0402FR-07100RL; 100 Ω, 1 %, 0402	60	0.58	34.80	Bias network
Resistor	Yageo RC0402FR-07100KL; 100 kΩ, 1 %, 0402	60	0.37	22.20	Bias network
Capacitor	Taiyo Yuden TMK107B7105KA-T; 1 µF, 25 V, X7R, 0603	50	1.55	77.50	Supply decoupling
Capacitor	Murata GCM155R71H104KE02J; 100 nF, 50 V, X7R, 0402	60	0.50	30.00	Supply decoupling
Laminate	FR4, double-sided copper, 100 × 75 × 1.5 mm	2	72.00	144.00	PCB stock
ESP32 board	ESP32-WROOM-32, 38-pin development board	2	449.00	898.00	Wi-Fi TX and RX
Tweezers	Straight ESD-safe tweezers (ESD-11)	1	35.00	35.00	SMD assembly
Controller	Arduino board, model to be confirmed	—	—	—	Diode state control
Total of listed line items				**Rs 2834.50**	

6.5 Component selection and bill of materials

Table 6.2 lists the components bought for the prototype.

PIN diode. The BAR63-03W was chosen for its low capacitance and low forward resistance at a few milliamperes. The order record quotes 0.21 pF and 1 Ω; these are datasheet figures at specific conditions (Table 3.2), not the equivalent-circuit values of the HFSS model, which are not recorded. The OFF-state value relevant to this bias circuit is the zero-bias 0.30 pF.

RF choke. The 0402HP-22N has a high self-resonant frequency for its inductance and a DC resistance low enough to be negligible in (6.1). Its limitation at 2.45 GHz is that it is close enough to self-resonance for its effective inductance to be well above 22 nH.

MOSFET, resistors and capacitors. The AO3401A matches the SOT-23 land pattern and switches fully with 3.3 V gate drive. The three resistor values correspond, by the analysis of Section 6.3, to the series current limiter (220 Ω), the gate pull-up (100 kΩ) and the gate resistor (100 Ω).

ESP32 boards. The ESP32-WROOM-32 module operates from 2412 MHz to 2484 MHz, transmits up to 19.5 dBm (802.11b) and has a typical sensitivity of −97 dBm at 1 Mbps, with an on-board PCB antenna. Its I/O operates at 3.3 V.

6.6 Fabrication constraints and the deferral of the 4×4 board

The unit cell was chosen as the first physical prototype for practical reasons.

- **Size.** The 4×4 board is 300 mm square. It cannot be made from desktop PCB mills.
- **Cost.** Outsourced fabrication of a board of this size was beyond the project budget at the time of writing; the team has therefore deferred it.
- **Assembly.** The array needs 16 diodes, 32 chokes, 16 transistors, 48 resistors and 32 capacitors, mostly 0402, soldered by hand.
- **Risk.** A single cell tests the assembly process, the bias circuit, the controller interface and the measurement chain at low cost. Problems found there can be corrected before the array is built.

6.7 Proposed hardware architecture

Figure 6.5 shows the planned test arrangement. Two ESP32-WROOM-32 boards form the Wi-Fi link: one transmits packets and the other receives them and logs the RSSI of each. The unit-cell PCB is placed so that it reflects part of the transmitted signal towards the receiver. An Arduino board switches the diode through the on-board driver and records the state changes. The Wi-Fi link and the control wiring are separate: the Arduino does not take part in the radio link.Table 6.3 lists the planned connections.

Wi-Fi channel. The channel should be fixed for the whole experiment and chosen close to the 2.45 GHz design frequency. Channels 8 and 9 are centred at 2447 and 2452 MHz [1], within 3 MHz of it. The occupancy of the chosen channel by other networks should be recorded, because interference affects packet reception.

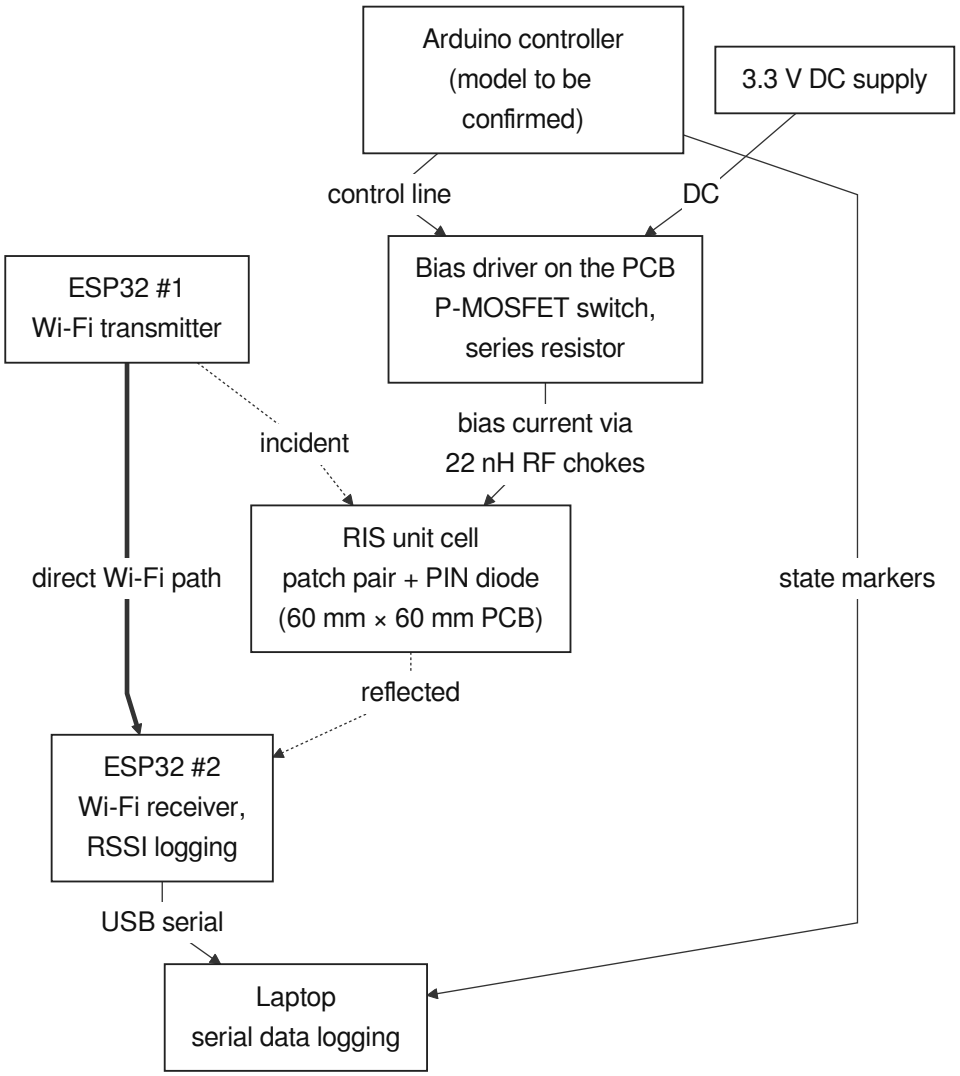
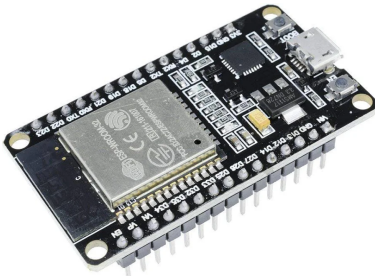


Figure 6.5. Proposed hardware architecture.

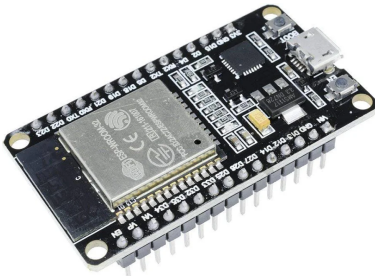
Table 6.3. Planned connections of the proposed test hardware.

From	To	Signal	Notes
Arduino GPIO (pin TBC)	PCB pad J1	Diode control, active low	Switches Q1 with 3.3 V or 5 V logic in the inferred circuit
3.3 V supply (source TBC)	PCB pad J2	Bias supply	About 11 mA when ON
Arduino and supply ground	PCB rear copper	Common ground	No top-side ground pad
Arduino GPIO (pin TBC)	ESP32 RX GPIO (pin TBC)	State marker (optional)	Level-shift if the Arduino uses 5 V logic; ESP32 I/O is 3.3 V
ESP32 RX	Laptop	USB serial	Per-packet log
Arduino	Laptop	USB serial	State-change log

6.8 Hardware photographs



(a) ESP32 transmitter board

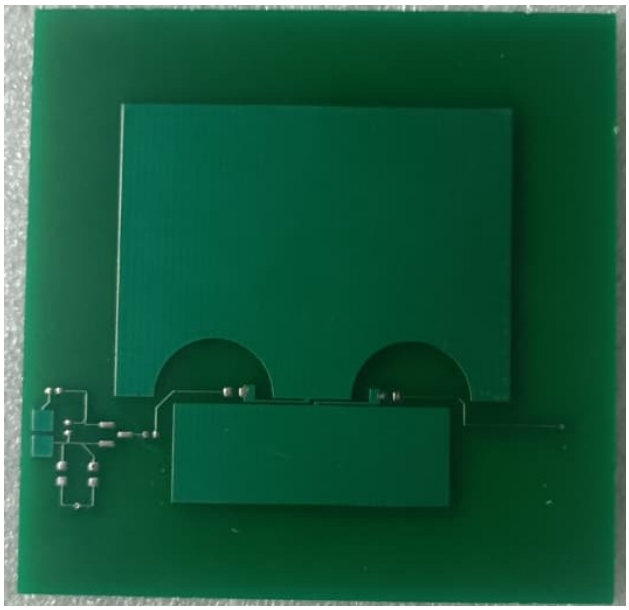


(b) ESP32 receiver board

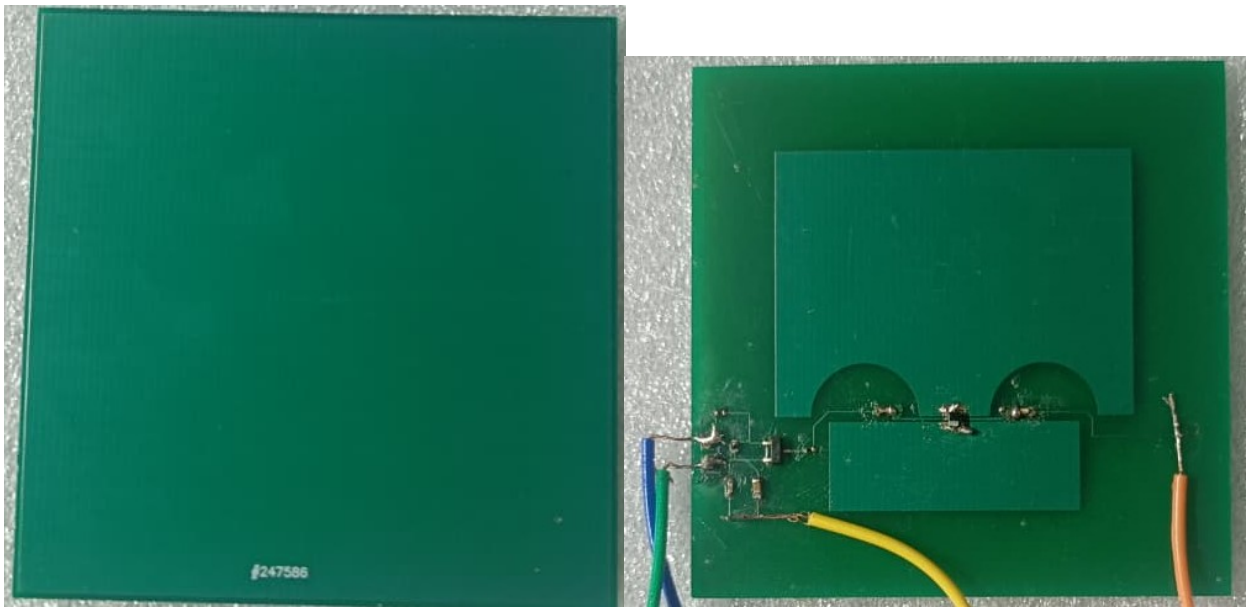


(c) Arduino controller

Figure 6.6. Photographs of the test boards.



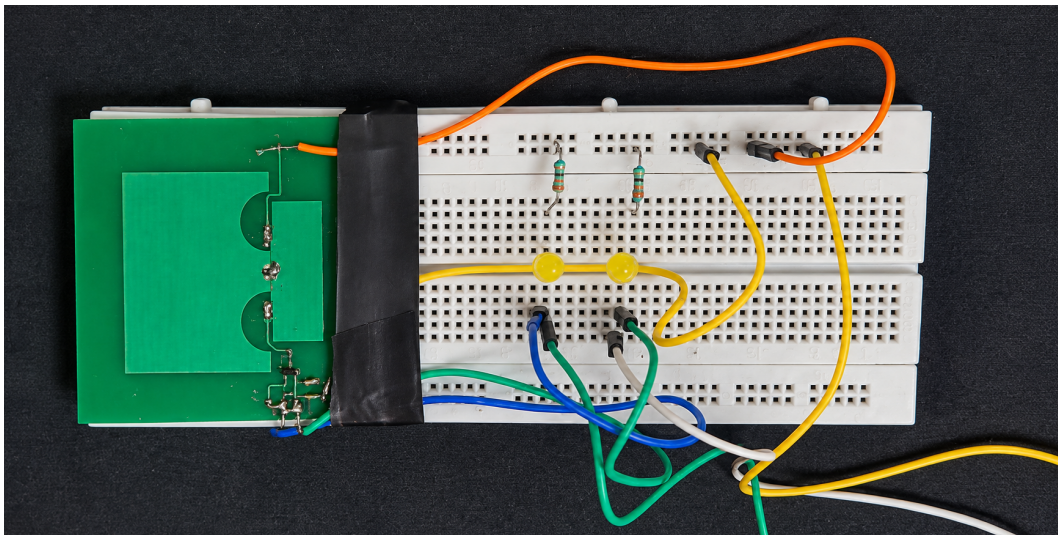
(a) Fabricated unit-cell PCB, front



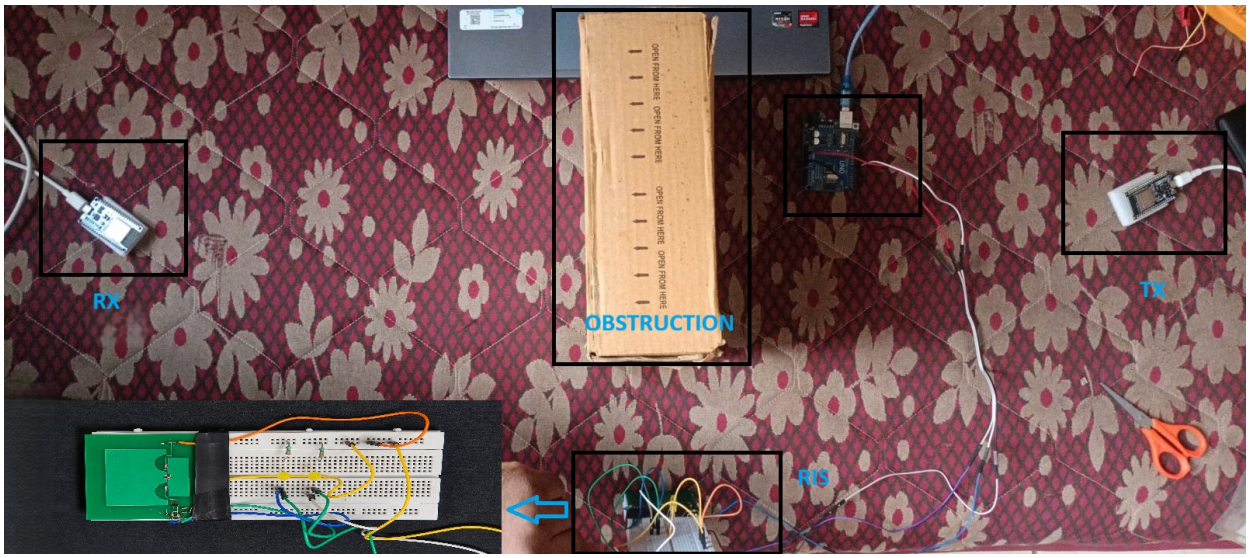
(b) Unit-cell PCB, rear

(c) Assembled unit cell with mounted components

Figure 6.7. RIS manufacuted PCB



(a) RIS Assembly



(b) Complete TX–RIS–RX arrangement

Figure 6.8. Assembled unit cell and the proposed TX–RIS–RX test arrangement.

CHAPTER 7

EXPERIMENTAL METHODOLOGY, RESULTS

7.1 Experimental Methodology

7.1.1 Objective of the Hardware Experiment

The objective of the hardware experiment was to investigate whether the fabricated Reconfigurable Intelligent Surface (RIS) unit cell could modify the received Wi-Fi signal strength in an indoor environment in the presence of an obstruction. The Esp 32 was used as TX and RX and four experimental conditions were considered for evaluation: an unobstructed baseline, an obstructed path without the RIS, an obstructed path with the RIS in the OFF state, and an obstructed path with the RIS in the ON state. Measurements were collected at transmitter–receiver separations of 60 cm, 1 m, and 3 m as shown in Figures 7.1, 7.2, and 7.3. The primary performance metric was the change in mean RSSI relative to the obstructed condition without the RIS. A positive difference indicates stronger received signal strength, whereas a negative difference indicates weaker received signal strength. The RIS-OFF versus RIS-ON comparison was also retained to evaluate the difference between the two operating states.

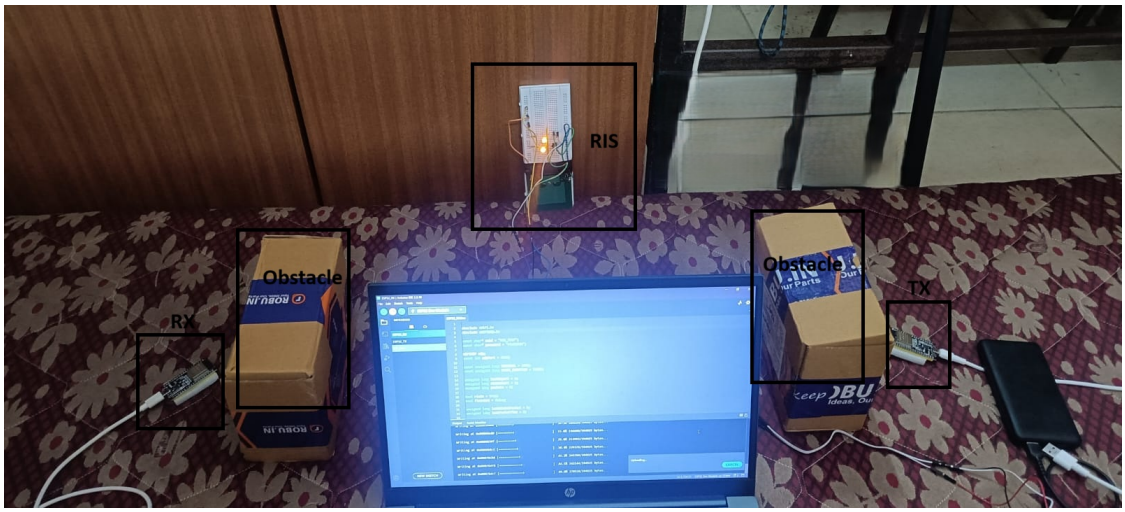


Figure 7.1. Experimental setup at 60 cm.

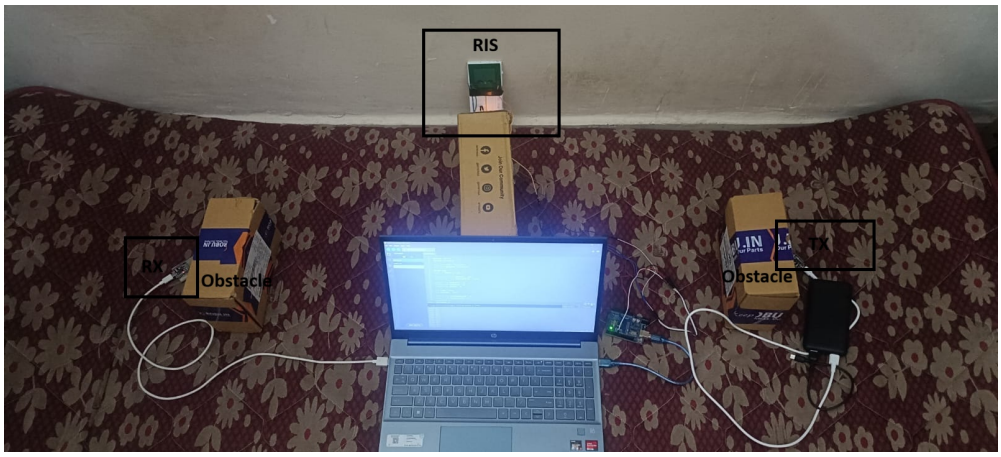


Figure 7.2. Experimental setup at 1 m.



Figure 7.3. Experimental setup at 3 m.

7.1.2 Measurement Procedure

The experimental setup consists of a ESP32-based Wi-Fi transmitter, a ESP32 receiver, the fabricated RIS unit cell, and a indoor propagation environment. The receiver logges RSSI readings and received-packet counts during the measurement process which is collected and then exported to MATLAB for analysis. The following steps were followed during the experiment:

- **Unobstructed baseline:** The received signal strength was recorded without the obstruction and without the RIS panel in the propagation path.
- **Obstructed condition:** An obstruction was introduced into the direct propagation path, and the received signal strength was recorded with the RIS absent.
- **RIS-OFF condition:** The RIS panel was installed and measurements were collected with the RIS in the OFF state.
- **RIS-ON condition:** The obstruction and panel arrangement was perserved while the RIS was switched to the ON state. The resulting RSSI measurements were recorded for comparison with the other conditions.
- **Distance variation:** The measurements were performed at transmitter–receiver distances of 60 cm, 1 m, and 3 m.
- **Data processing:** The recorded logs were processed in MATLAB. The mean, median, standard deviation, and number of valid samples were calculated for each distance and experimental condition.

7.2 Hardware Experimental Results

7.2.1 Mean RSSI Under the Four Experimental Conditions

Evaluation of the mean RSSI under all four experimental conditions at the three transmitter–receiver distances. Figure 7.4 presents the resulting comparison.

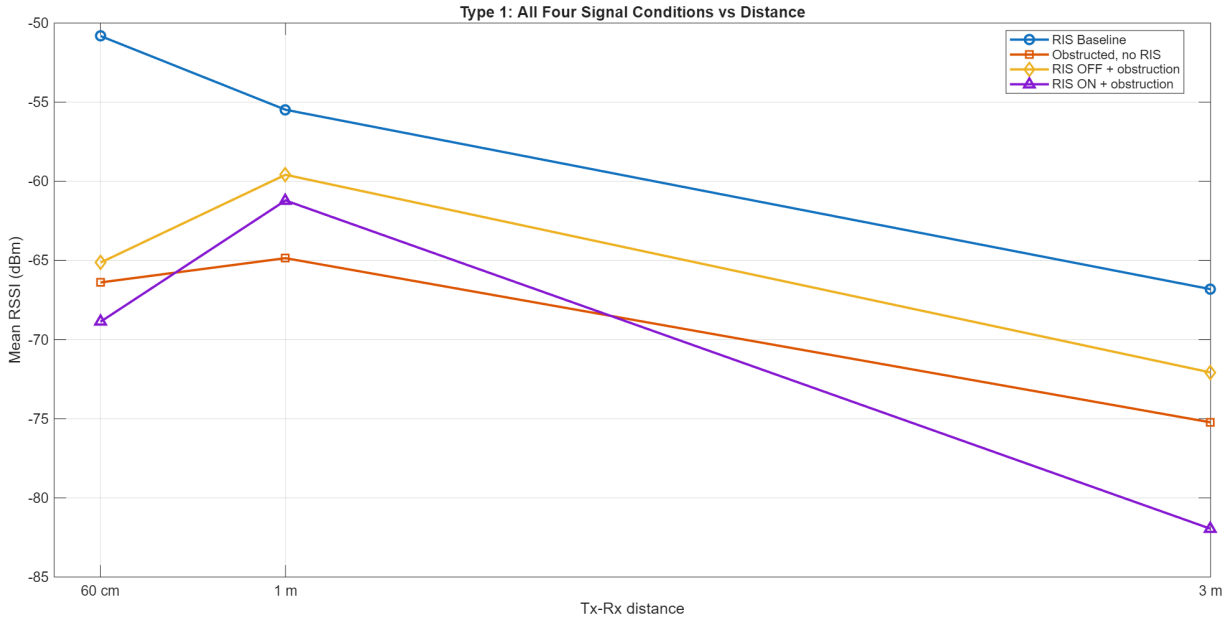


Figure 7.4. Mean RSSI for the unobstructed baseline, obstructed path without RIS, RIS-OFF state, and RIS-ON state at different transmitter–receiver distances.

At 60 cm, the unobstructed baseline produced a mean RSSI of approximately −50.82 dBm, whereas the obstructed condition without the RIS produced approximately −66.39 dBm. Also it was obsever that at 1 m, the corresponding means were −55.48 dBm and −64.85 dBm and at 3 m, they were observed to be −66.81 dBm and −75.23 dBm, respectively.

These results lead us to understand that the obstructed measurements had lower mean RSSI than the corresponding unobstructed baselines at all three tested distances. The differences between the baseline and obstructed means was calculated to be approximately 15.57 dB, 9.38 dB, and 8.41 dB at 60 cm, 1 m, and 3 m, respectively.

7.2.2 Obstructed Condition Compared with RIS OFF and RIS ON

Evaluation to addresses whether the measured signal strength increased when the RIS panel was introduced and activated. Figure 7.5 compares the mean RSSI for the obstructed/no-RIS, RIS-OFF, and RIS-ON conditions.

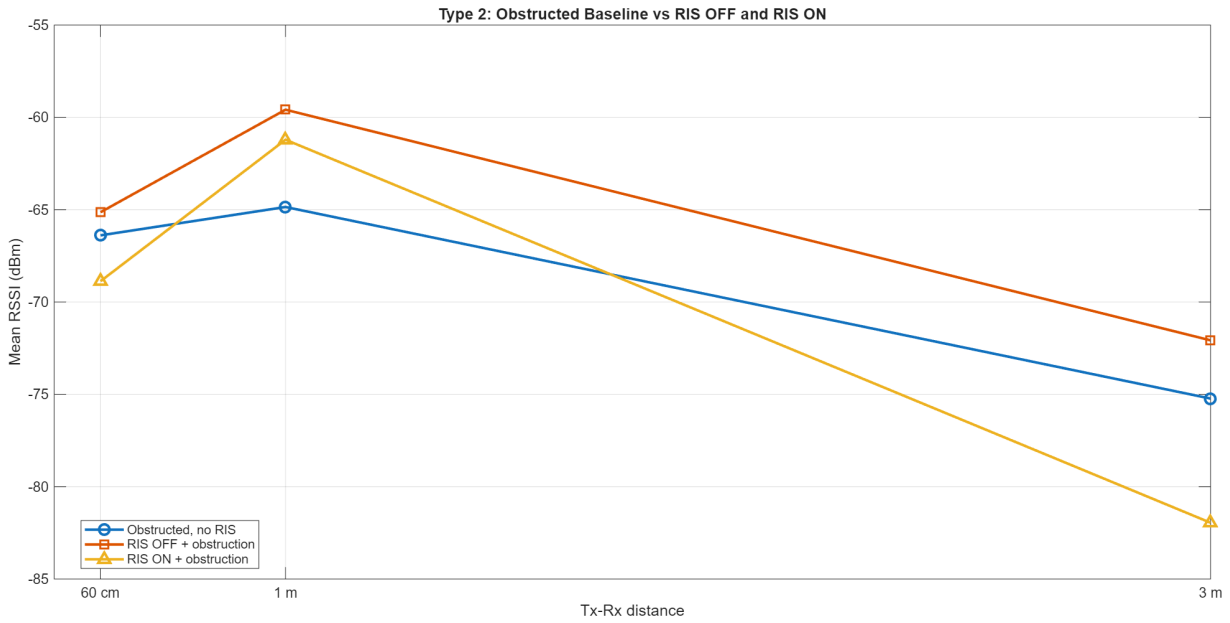


Figure 7.5. Mean RSSI for the obstructed condition without RIS and with the RIS in the OFF and ON states.

To quantify the variation in the RSSI change relative to the obstructed/no-RIS condition was calculated as

$$\Delta RSSI_{\text{state}} = \overline{RSSI}_{\text{state}} - \overline{RSSI}_{\text{obstructed,no RIS}}. \tag{7.1}$$

Figure 7.6 presents these differences for both RIS states.

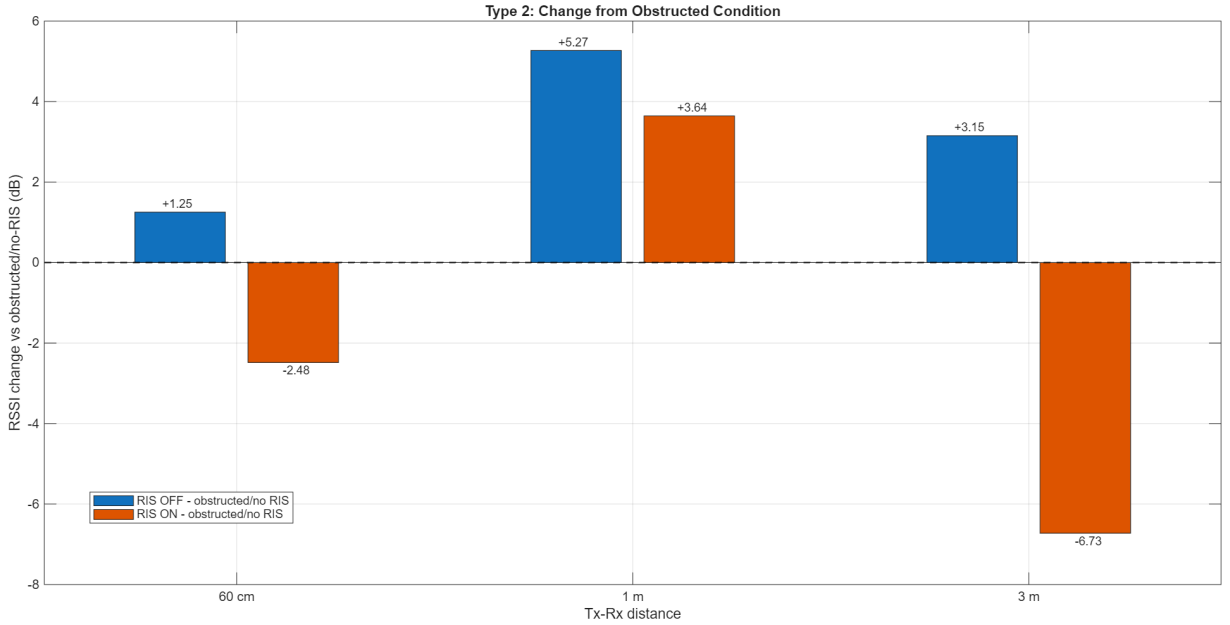


Figure 7.6. Change in mean RSSI for RIS OFF and RIS ON relative to the obstructed/no-RIS condition.

Table 7.1. Mean RSSI and change relative to the obstructed/no-RIS condition.

Distance	Obstructed (dBm)	RIS OFF (dBm)	OFF change (dB)	RIS ON (dBm)	ON change (dB)
60 cm	−66.39	−65.14	+1.25	−68.87	−2.48
1 m	−64.85	−59.58	+5.27	−61.21	+3.64
3 m	−75.23	−72.07	+3.15	−81.95	−6.73

At 60 cm, the RIS-OFF condition showed an increase of approximately 1.25 dB relative to the obstructed/no-RIS condition. The RIS-ON condition, however, showed a decrease of approximately 2.48 dB. Similarly for 1 m, both RIS states produced higher mean RSSI than the obstructed/no-RIS condition. The RIS-OFF condition showed an increase of approximately 5.27 dB, while the RIS-ON condition showed an increase of approximately 3.64 dB. This was the only tested distance at which the RIS-ON condition exhibited a positive RSSI change relative to the obstructed/no-RIS reference. At 3 m, the RIS-OFF condition showed an increase of approximately 3.15 dB, whereas the RIS-ON condition showed a decrease of approximately 6.73 dB relative to the obstructed/no-RIS measurement.

The measurements allow us to come to the conclusion that the RIS-ON state did not produce a consistent increase in RSSI across all tested distances. The positive change observed at 1 m is a preliminary result that requires repeat measurements under controlled conditions.The RIS-OFF measurements also produced higher mean RSSI than the obstructed/no-RIS reference at all three distances. This suggests that the presence of the panel may alter the propagation environment through passive reflection, scattering, or other interactions with the local multipath channel.

7.2.3 Summary of RSSI Statistics

Table 7.2 summarizes the mean and standard deviation of the recorded RSSI samples.

Table 7.2. Summary statistics of the recorded RSSI measurements.

Distance	Condition	Mean (dBm)	SD (dB)	Samples
60 cm	Baseline	−50.82	0.94	110
	Obstructed, no RIS	−66.39	3.72	108
	RIS OFF	−65.14	1.32	110
	RIS ON	−68.87	2.10	110
1 m	Baseline	−55.48	1.29	109
	Obstructed, no RIS	−64.85	1.89	109
	RIS OFF	−59.58	1.21	108
	RIS ON	−61.21	2.33	110
3 m	Baseline	−66.81	1.37	107
	Obstructed, no RIS	−75.23	1.54	106
	RIS OFF	−72.07	2.30	108
	RIS ON	−81.95	2.28	103

7.3 Limitations

The measurements give us some idea of the performance of the fabricated RIS unit cell in an indoor Wi-Fi environment. We've taken the measurements at a range of distances and under a range of obstruction conditions so we can't really isolate the effect of distance. The data we have is also not sufficient to establish the statistical significance of the differences between the RIS states. So there is a need to do repeated measurements under the same conditions to assess the consistency of the results and to quantify the experimental variability. The RIS-ON state didn't consistently perform better than the obstructed, no-RIS reference. The received signal appears to be a function of a number of factors including panel orientation, diode bias, reflection phase and the relative positions of Tx, RIS and Rx. Future work should look at these parameters in a systematic way with the measurement geometry held constant. It would also be useful to separate out the contribution of passive reflection from the panel from the effect of switching the diode state in order to better understand the signal changes we are observing.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project examined the design, simulation, fabrication, and initial testing of a PIN diode-controlled Reconfigurable Intelligent Surface (RIS) at 2.45 GHz. It involved HFSS electromagnetic simulation, MATLAB array analysis, PCB fabrication and ESP32 based Wi-Fi testing. The unit cell simulations showed a reflection phase difference of around 140° between the diode states. Reflection losses (particularly in the OFF state) are still a constraint. The simulations also showed that the bias network has an effect on the EM response and must be taken into account during the optimisation process. MATLAB simulations of the 2×2 and 4×4 arrays demonstrated the potential for reflected beam control via one bit of phase switching. The larger array gave a more directive response, though phase quantisation did limit the achievable steering accuracy. These simulation results need to be verified experimentally. A single RIS unit cell was fabricated and tested using an ESP32 based Wi-Fi link. Measurements were taken at 60 cm, 1 m and 3 m under both unobstructed and obstructed conditions with the RIS in both the ON and OFF states. At 1 m the RIS-ON condition improved the RSSI by approximately 3.64 dB over the obstructed link without the RIS. The signal strength was worse at the other two distances. The RIS-OFF condition gave a higher mean RSSI than the obstructed/no RIS condition in all three measurements.

Overall it establishes a practical workflow from unit cell design and EM simulation through to PCB fabrication and wireless testing. It also identifies reflection loss, bias network design and measurement repeatability as the areas that need further work.

8.2 Future Work

The next stage of the project will focus on the following tasks:

- **Improve the unit cell:** Verify the PCB footprints and component values, and optimise the diode and RF choke models to improve reflection efficiency and phase separation.
- **Repeat the hardware measurements:** Keep the transmitter, receiver, obstruction, and RIS geometry fixed. Repeat the ON/OFF tests to determine whether the measured changes are consistent.
- **Build a 4×4 array:** Correct the array pitch and PCB layout, integrate the control circuitry, and test the array's reflection and beam-steering performance.
- **Compare array sizes:** Measure the received signal at different distances using arrays of different sizes and compare the results with MATLAB predictions.
- **Develop adaptive control:** Investigate using receiver RSSI feedback to select RIS configurations that improve the received signal under different indoor conditions.

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